

JAIBB

Hand Notes

On

Principles of Economics (POE)

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Module-A: Economics-Introduction

1. Write down the features of Micro Economics and Macro Economics. 96(1.b)

Features of Microeconomics and Macroeconomics

Microeconomics:

Definition: Microeconomics analyzes the economic activities and decision-making processes of individual economic units, such as households and businesses.

Features:

- **Small-scale analysis:** Focuses on individual and business decision-making.
- **Price determination:** Prices are set based on market demand and supply.
- **Consumer behavior:** Studies how demand for goods and services is influenced by various factors.

Macroeconomics:

Definition: Macroeconomics examines the overall economic activities of a country or national economy.

Features:

- **Large-scale analysis:** Focuses on national income, employment, inflation, etc.
- **Economic policies:** Analyzes the effectiveness of government economic policies and their impacts.
- **Aggregate demand and supply:** Determines the relationship between total demand and supply in the economy.

2. Discuss the differences between microeconomics and macroeconomics. 99(1.a)

Differences between Microeconomics and Macroeconomics

Microeconomics and Macroeconomics are the two main branches of economics. Their key differences are discussed below:

Aspect	Microeconomics	Macroeconomics
Definition	Microeconomics studies the economic behavior and decision-making of individuals or households. It analyzes market dynamics, demand and supply, price determination, and consumer-producer interactions.	Macroeconomics studies the overall economic activities and trends of a country or nation. It focuses on national income growth, inflation, employment, and economic development.

Scope	Covers business decisions, expenditures, demand, and consumer behavior.	Covers national income growth, inflation, and economic stability.
Objective	Helps in individual decision-making and understanding market trends.	Develops macroeconomic policies and ensures balanced national economic growth.
Level of Analysis	Micro-level (individual and firm-level analysis).	Macro-level (national and global economy analysis).
Models Used	Demand and supply models.	Macroeconomic models like national income measurement.
Key Focus Areas	Consumer choices, production functions, costs, and profits.	Economic growth, employment, and economic policies.

3. "Microeconomics and Macroeconomics are interdependent and complement each other" Explain the notion by describing the features of both Micro and Macroeconomics. 98(1.b)

Microeconomics and Macroeconomics Are Interdependent and Complement Each Other

Microeconomics and Macroeconomics are two major branches of economics that are closely related and influence each other. Their interdependence can be explained through the following points:

Microeconomics:

- It focuses on specific and individual decision-making processes.
- Analyzes economic activities of individuals, households, or firms.
- **Example:** A business owner's decisions on pricing, production, and earnings.

Macroeconomics:

- Examines economic issues from a broad perspective.
- Studies national, regional, and global economic factors.
- **Example:** Unemployment rate, GDP growth, and inflation in a country.

Interdependence:

- Both micro and macroeconomics influence each other.
- **Example:** If inflation is controlled at the macro level, it directly affects business pricing and production at the micro level.
- Changes in microeconomic conditions, such as consumer demand, impact macroeconomic factors like national income and economic stability.

Conclusion:

Thus, microeconomics and macroeconomics are complementary. One's effectiveness depends on the other, forming a complete economic framework.

4. What are positive and normative economics? 97(1.a)

Positive and Normative Economics

Economics can be broadly divided into two branches: **Positive Economics** and **Normative Economics**.

Positive Economics:

- It deals with objective analysis based on facts and real data.
- It explains economic phenomena without any personal opinions or judgments.
- Uses statistical models and empirical research to analyze economic situations.
- **Example:** "Bangladesh's GDP grew by 5% last year." This is a positive economic statement as it is based on factual data and not influenced by opinions.

Normative Economics:

- It focuses on value judgments, ethics, and what ought to be done in an economy.
- It suggests policies or actions based on societal welfare and moral considerations.
- Does not rely purely on factual analysis but includes opinions on what is desirable.
- **Example:** "The government should strengthen social security programs for the poor." This is a normative economic statement as it expresses an opinion about what should be done.

Thus, **positive economics** explains "what is" based on facts, while **normative economics** suggests "what ought to be" based on value judgments.

5. Differentiate between positive and normative economics. 99(1.c)

Differences between Positive and Normative Economics:

Aspect	Positive Economics	Normative Economics
Definition	Analyzes the economy based on real data and observations. It explains what is happening and why it is happening.	Evaluates the economy based on opinions and value judgments. It discusses what ought to happen.
Objective	Explains events and provides fact-based information.	Focuses on policy-making and moral values.
Example	"The national income of the country has increased."	"The national income should be increased."
Focus Area	Data analysis, statistics.	Ethics, policies, and social issues.

Thus, **positive economics** is fact-based and descriptive, whereas **normative economics** is opinion-based and prescriptive.

"Scarcity Brings in the Problem of Choice" – Explanation

Definition of Scarcity:

Scarcity refers to a situation where resources, goods, or services are limited compared to human wants and needs. Since resources are not unlimited, people must make choices about how to allocate them efficiently.

How Scarcity Leads to the Problem of Choice:

1. Limited Availability of Resources:

- ✓ Since resources like land, labor, and capital are scarce, individuals and businesses must decide how to use them effectively.
- ✓ Example: A farmer with limited land must choose whether to grow rice or wheat.

2. Decision-Making Pressure:

- ✓ Consumers must prioritize their needs and wants based on limited income and available goods.
- ✓ Example: A person with a fixed budget must choose between buying a mobile phone or a laptop.

3. Increase in Prices:

- ✓ Scarcity often leads to higher demand for certain goods, increasing their prices and making choices more difficult.
- ✓ Example: During a crisis, the price of essential goods like food and fuel rises, forcing consumers to reconsider their spending.

4. Trade-Offs and Opportunity Cost:

- ✓ Every choice involves an opportunity cost, which is the next best alternative forgone when making a decision.
- ✓ Example: If a government spends more on defense, it may have to reduce spending on education.

5. Psychological and Economic Impact:

- ✓ Scarcity can create stress among consumers, as they struggle to make the best possible decision.
- ✓ Businesses may also face challenges in meeting demand due to resource limitations.

Conclusion:

Since human wants are unlimited but resources are limited, scarcity forces individuals, businesses, and governments to make choices. This problem of choice influences economic decision-making and shapes how resources are allocated in a society.

7. What is an opportunity cost? Suppose a person holds cash BDT 1 lakh, which, if he places for a term deposit, will generate BDT 5,000 as interest. Instead, he decides to enroll in a professional course with that amount of money. What is the opportunity cost of the enrollment? Explain. 99(2.c)

Opportunity Cost:

Opportunity cost is the cost of choosing one alternative over another, which means the value of the next best alternative that is forgone.

Example:

Suppose a person has BDT 1 lakh. If he deposits this money in the bank, he will earn BDT 5,000 as interest. However, if he decides to spend the money on enrolling in a professional course, his opportunity cost will be:

Amount earned by depositing in the bank: **BDT 5,000** (which he would have earned if he didn't spend the money on the course).

Therefore, his opportunity cost is **BDT 5,000**.

Here is a **bar graph** showing the opportunity cost:



- **Fixed Deposit (Earn 5,000 BDT):** The opportunity of earning BDT 5,000 is shown (green bar).
- **Professional Course (Lost 5,000 BDT):** The opportunity cost of losing this BDT 5,000 is shown (red bar).

This **graph clearly shows** that by investing in the professional course, the person is forgoing the interest income of BDT 5,000, which represents the **opportunity cost**.

8. What is the opportunity cost? Explain the relevance of opportunity cost in economics with example.

Opportunity Cost:

Opportunity cost refers to the value or benefit that must be given up when choosing one alternative over another. In simple terms, when we make a decision, the most valuable alternative we forgo is the opportunity cost. It plays an important role in our decision-making process.

Relevance of Opportunity Cost in Economics:

1. **Making the Right Decision:**

Understanding opportunity cost ensures that we make the best use of our limited resources. It

allows us to compare different alternatives and helps us make more informed decisions.

2. **Resource Management:**

Opportunity cost teaches us how to manage our resources effectively. When one resource or product is chosen, other possible opportunities are forgone, which impacts our economic activities.

3. **Economic Growth:**

When businesses or governments make decisions about projects or investments, analyzing opportunity costs helps them understand which project will yield the maximum return. This allows them to make decisions that contribute to economic growth.

4. **Market Efficiency:**

Opportunity cost is a key part of market efficiency. It plays a significant role in the decision-making process between buyers and sellers. When a buyer chooses a product, they consider the opportunity cost of other products they are forgoing.

Example:

Suppose a student has to decide between two options during his summer vacation: joining a job or enrolling in a university course. If he chooses the job, he will lose the opportunity to learn from the course, and the value of what he could have learned becomes his opportunity cost.

Conclusion:

Opportunity cost is an essential concept in economics. It shapes our decision-making process and helps us use resources effectively. By understanding opportunity cost, we can evaluate different aspects of our lives and economic activities more accurately.

9. What are the economic resources that we use to produce goods and services? Explain. 97(1.c)

Four Major Economic Factors for Production of Goods and Services

1. **Land:**

- ✓ **Definition:** Land refers to natural resources used in all types of production. It includes agricultural land, minerals, and other natural resources.
- ✓ **Explanation:** Land plays a vital role in agriculture, industrial production, and construction. The quality and availability of land affect productivity.

2. **Labor:**

- ✓ **Definition:** Labor refers to human physical and mental efforts required for producing goods and services.
- ✓ **Explanation:** Labor has different skill levels. Skilled workers enhance productivity, while unskilled workers may have lower efficiency. Education, training, and experience help improve the quality of production.

3. Capital:

- ✓ **Definition:** Capital includes machinery, tools, and technology used in the production process.
- ✓ **Explanation:** Examples of capital include factory machines, production technology, and office equipment. The availability of capital makes production faster and more efficient.

4. Entrepreneurship:

- ✓ **Definition:** Entrepreneurs are individuals who combine resources to create new goods and services while taking risks.
- ✓ **Explanation:** Entrepreneurs introduce new ideas, respond to market demands, and adapt to technological changes. They play a crucial role in economic growth by bringing innovative products to the market.

These four factors work together to produce goods and services, forming the foundation of an efficient economic system.

10. What is Production Possibilities Curve (PPC)? What happens to PPC frontier of an economy if advanced technical education improves the overall skills and knowledge of human capital? Explain with diagram.
98(4.c)

Production Possibility Curve (PPC)

The **Production Possibility Curve (PPC)** is a graphical representation that shows the possible production combinations of two goods or services in an economy using limited resources. PPC is a concave curve that illustrates how an economy can maximize the production of two different products with available resources.

Key Features of PPC:

- **Production Limit:** PPC determines how increasing the production of one good affects the reduction of another good due to resource constraints.
- **Inefficiency:** If production is below the PPC, it indicates inefficient use of resources.
- **Maximum Efficiency:** Points on the PPC represent the most efficient use of resources.

Impact of Technological Advancement on PPC

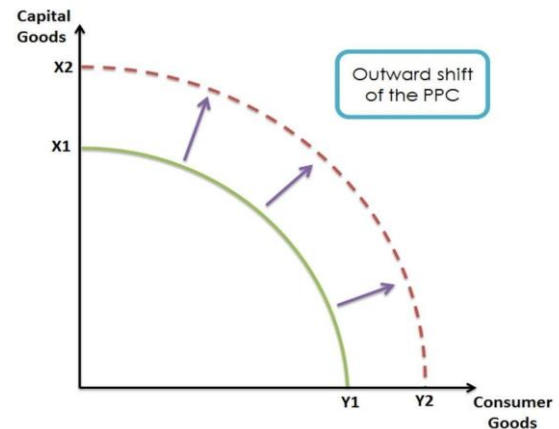
Improved technological education enhances human resource skills and boosts production capacity. The impact of technological advancement on PPC includes:

1. **Shift (Expansion):** As technology improves, production efficiency increases, shifting the PPC outward. This indicates that the economy can produce more goods.

2. **New Technology:** Automation, advanced machinery, and better production methods reduce costs and increase output.

Graphical Explanation

- **Initial PPC (X1):** Represents the maximum production limits of two goods. When resources are used efficiently, production occurs on this curve.
- **Shift Due to Technological Advancement:** With better technology, PPC shifts outward (X2), indicating that the economy can produce more goods using the same resources.



Conclusion

PPC is an essential economic model that represents production capacity and limitations. Technological education and innovation improve resource efficiency, shifting PPC outward and leading to higher economic growth.

11. "Economic models are used to test economic theories"-Explain the notion with example. 97(1.b)

Economic Model

An economic model is an analytical framework used to understand the behavior of real economic events. It combines fundamental economic concepts, theories, and equations to analyze the impact of economic policies and changes.

Examples:

1. **Supply and Demand Model:**

- This model explains the relationship between the price and quantity of a product.
- **Key Concept:**
 - ✓ When **demand increases**, the **price of the product also rises**.
 - ✓ When **supply decreases**, the **price may rise even further**.
- **Example:**
 - If the government increases the price of rice, **demand will decrease**, as consumers will look for cheaper alternatives.
 - A new **equilibrium price** will be established in the market.

2. **Monetary Policy Model:**

- This model explains how **the central bank influences the economy by changing interest rates**.
- **Key Concept:**

- ✓ **Higher interest rates** increase the **cost of borrowing**, leading to **lower investments** and reduced economic activity.
- ✓ **Lower interest rates** make borrowing easier, leading to **higher investments and spending**.

○ **Example:**

- ✓ If Bangladesh Bank **raises interest rates**, the cost of taking loans from banks will increase, reducing business expansion.

Conclusion:

Economic models play a crucial role in assessing real economic situations and making forecasts. They help policymakers understand economic changes and make informed decisions.

12. What is the circular flow model in economics? What are the elements on sectors that make up a circular flow? Explain. 98(1.c)

Circular Flow Model

The Circular Flow Model is a concept that illustrates how money, goods, and services flow between households, businesses, government, and foreign entities in an economy.

Components/Sectors of the Model:

1. **Households:**

- ✓ Households act as both **producers and consumers**.
- ✓ They supply **labor** to businesses and use the earned income to purchase goods and services.

2. **Businesses:**

- ✓ Businesses **produce goods and services** and purchase labor and other inputs from households.
- ✓ They pay wages to workers and sell products to consumers.

3. **Government:**

- ✓ The government collects **taxes** and provides **public goods and services** such as infrastructure, education, and healthcare.

4. **Foreign Sector:**

- ✓ It represents economic relations with foreign markets, including **exports (selling goods abroad)** and **imports (buying goods from other countries)**.

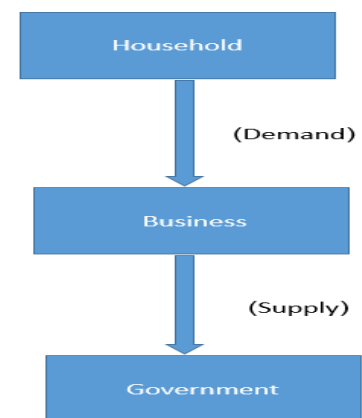


Figure: Circular Flow Model

Key Aspects:

- **Flow of Money:** Households pay businesses for goods and services, while businesses pay households in the form of wages and salaries.
- **Flow of Goods:** Businesses produce goods for households, and households purchase these goods.

Conclusion:

The Circular Flow Model helps to understand the **interdependence** between different sectors of the economy. It provides a **comprehensive view** of how economic activities function within a society.

13. What is Market Economy and Regulated Economy? Between them which one is better for economic development? 96(3.c)

Market Economy:

A market economy is an autonomous economic system where prices and production of goods and services are determined by **demand and supply** in the market. It operates based on **consumer and producer freedom** and encourages **competition**.

Controlled Economy:

A controlled economy is a system where the **government regulates** various economic aspects such as **pricing, production levels, and resource allocation**. In this system, the government's **intervention is significant** in economic activities.

Which is better for Economic Development?

✓ **Market Economy:**

- More **efficient and innovative** as it encourages **competition and entrepreneurship**.
- Leads to **higher productivity and economic growth**.

✓ **Controlled Economy:**

- Helps to **address market failures** and **social issues** such as income inequality.
- However, **excessive control** can **hinder development** and reduce efficiency.

Conclusion:

A **market economy is generally more beneficial** for economic development. However, a **balanced approach** that combines **government regulation with market efficiency** can be the most effective strategy for sustainable economic growth.

Module-B: Basics of Microeconomics

1. Explain the law of Supply. Explain the factor for changes in supply.

Law of Supply

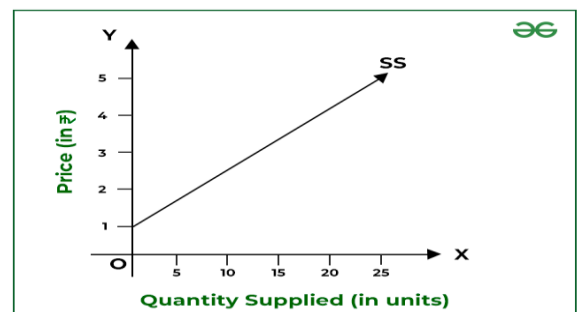
The **Law of Supply** states that **when the price of a product increases, its supply also increases, and when the price decreases, the supply decreases**, assuming **other factors remain constant (Ceteris Paribus)**.

Key Reason:

Producers aim to **maximize profit**—they supply more when prices are high and reduce supply when prices are low.

Example:

Price (Taka)	Supply (Units)
1	5
2	10
3	15
4	20
5	25



Factors Affecting Supply (Other than Price):

1. Change in Production Cost:

- ✓ Higher costs (raw materials, labor, and technology) **reduce supply**.
- ✓ Lower costs **increase supply**.

2. Technological Advancements:

- ✓ Improved technology **reduces production costs** and **increases supply**.

3. Government Policies:

- ✓ **Taxes and tariffs** may **reduce supply**.
- ✓ **Subsidies** lower production costs and **increase supply**.

4. Natural Disasters & Environmental Factors:

- ✓ Floods, droughts, or earthquakes can **reduce agricultural supply**.

5. Market Competition & Number of Suppliers:

- ✓ More producers in the market **increase supply**.
- ✓ Fewer producers **reduce supply**.

6. Future Price Expectations:

- ✓ If producers expect prices to **rise**, they **reduce current supply**.
- ✓ If they expect prices to **fall**, they **increase current supply**.

Thus, supply is influenced by **price changes** as well as **several external factors** affecting production and market conditions.

2. Explain the concept of "change in supply" and "change in quantity supplied" using graph. 98(1.c)

Change in Supply and Change in Quantity Supplied

Change in Supply:

An increase or decrease in supply at the same price level is called a change in supply. It occurs due to external factors such as production costs, technological advancements, government policies, weather conditions, etc.

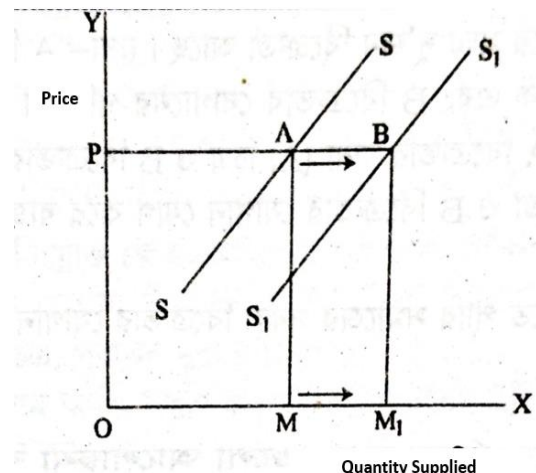
Graph of Change in Supply:

- When supply increases, the SS curve shifts to the right.
- When supply decreases, the SS curve shifts to the left.

In the graph, at price **OP**, the seller is at **point A on the SS curve**, supplying **Oq** quantity. Later, even though the price remains the same (**OP**), the seller moves to **point B on the new supply curve S_1S_1** , supplying **Oq₁** quantity. This occurs due to **reduced production costs, technological improvements, or government subsidies**. When the quantity supplied increases **without a price change**, it is called **Increase in Supply**.

Change in Quantity Supplied:

A change in quantity supplied occurs due to **price changes**, keeping all other factors constant.

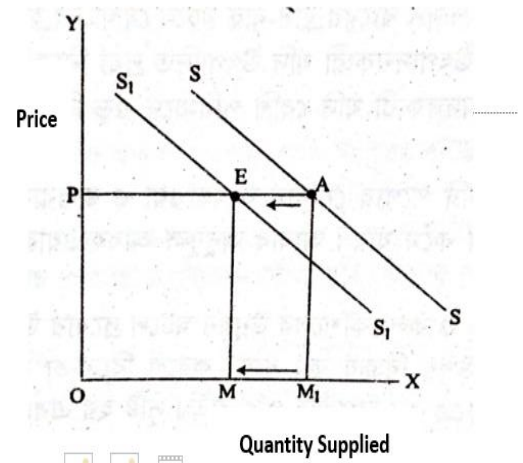


Graph of Change in Quantity Supplied:

- When the price increases from **P** to **P₁**, the quantity supplied expands from **Q** to **Q₁**.
- When the price decreases from **P₁** to **P**, the quantity supplied contracts from **Q₁** to **Q**.

Here, the quantity supplied moves **along the same supply curve (SS)**, from **E to E₁** and **B to C**.

- **Change in Supply** occurs due to external factors and shifts the **entire supply curve**.
- **Change in Quantity Supplied** occurs due to price changes and results in **movement along the same curve**.



This helps understand the dynamics of market supply and demand.

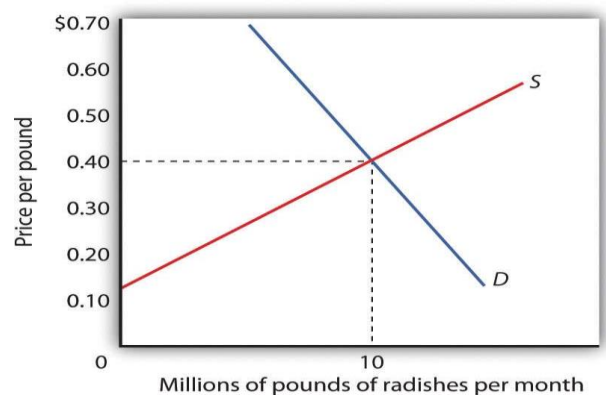
3. On a supply and demand diagram, show the equilibrium price, equilibrium quantity and the total revenue received by the producer. 97(4.c)

Demand and Supply Diagram

In the diagram below, **Equilibrium Price**, **Equilibrium Quantity**, and **Total Revenue** received by the producer are shown, indicating the relationship between demand (D) and supply (S).

Components:

1. **Equilibrium Price:**
The equilibrium price is determined at the price where the demand curve (D) and the supply curve (S) intersect. It is the price at which the quantity demanded equals the quantity supplied.
2. **Equilibrium Quantity:**
The equilibrium quantity is the quantity at which the demand curve (D) and the supply curve (S) meet. At this point, both the producer and the consumer are satisfied.
3. **Total Revenue:**
Total revenue is the product of equilibrium price (P) and equilibrium quantity (Q). That is,



$$\text{Total Revenue} = P \times Q$$

The demand and supply diagram helps determine market equilibrium. It clearly shows how equilibrium price and quantity are related and how the total revenue received by the producer is determined.

4. In monopoly there is no supply curve-Explain. 96(4.c)

The reasons for the absence of a supply curve in a monopoly are:

1. Single Seller:

In a monopoly, the sole seller controls the market price, which means there is no supply curve for them. As a result, there is no specific quantity supplied in the market.

2. Price Setting:

In a monopoly, the seller has the power to raise prices in the market. This keeps the supply curve unchanged because the seller fears that raising the price may reduce demand.

3. Barriers to Market Entry:

When new competitors are unable to enter the market, the seller maintains control over setting prices, which influences the supply curve.

4. Centralized Control:

The monopolist has complete control over the market for production, so their supply curve generally changes with the production quantity and market demand.

Therefore, in a monopoly, there is typically no conventional supply curve, and it is entirely influenced by the market demand.

5. Define the law of demand. Why does the demand curve slope downward? 99(2.a)

Law of Demand:

The Law of Demand is a fundamental principle in economics, which states that, holding everything else constant, when the price of a product increases, the demand for that product decreases, and when the price decreases, the demand increases. This means there is an inverse relationship between price and demand.

Slope of the Demand Curve:

The slope of the demand curve is downward because as the price increases, the attractiveness of purchasing that product decreases for consumers. For example, when the price of a product increases, consumers are more likely to purchase alternative products, leading to a decrease in demand.

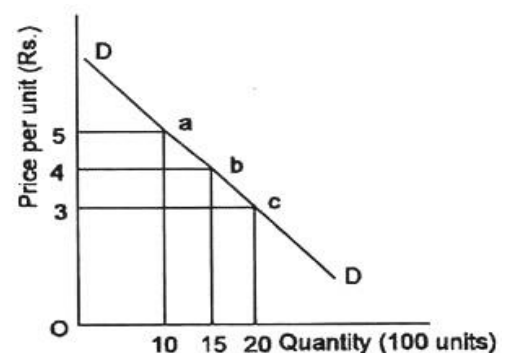


Fig. 1 : Downward-sloping Demand Curve

6. Explain the factor for changes in demand. 96(3.a)

Factor for changes in demand:

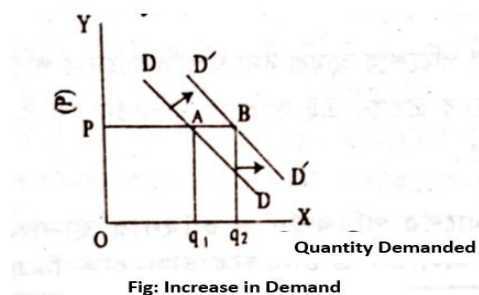
The various factors influencing demand changes, which affect consumer behavior in the market, are:

1. **Price Change:** When the price of a product increases, its demand generally decreases, and when the price decreases, the demand increases. This is explained by the "Law of Demand."
2. **Income Change:** When consumers' income increases, demand generally rises. For example, when people's income increases, they tend to buy luxury goods.
3. **Substitute Goods:** If two products are substitutes for each other, an increase in the price of one product may lead to an increase in the demand for the other product. For example, tea and coffee.
4. **Consumer Preferences and Tastes:** Changes in consumer preferences can affect demand. For example, if there is increased interest in healthy food, the demand for salads and fruits may rise.
5. **Population Change:** An increase in population generally increases demand because new consumers enter the market.
6. **Technological Advancements:** The discovery of new technology can increase the demand for some products. For example, the invention of smartphones may reduce the demand for traditional telephones.
7. **Marketing Activities:** Advertisements and promotions can increase consumer interest in a product, which helps to boost demand.

7. Explain the concept of "change in demand" and "change in quantity demanded" using graph. 98(2.b)

Change in Demand: Change in demand refers to the increase or decrease in demand at the same price. This happens due to changes in factors other than price, such as consumer income, preferences, market expectations, or changes in the perception of the product.

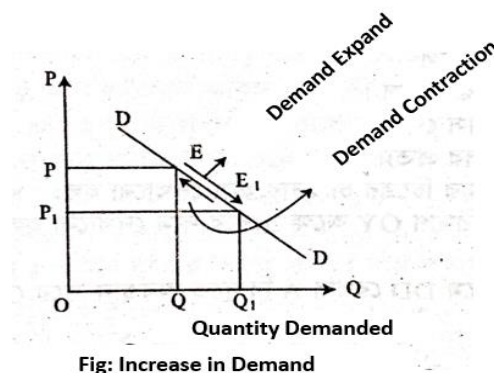
In the diagram, the increase in demand occurs when the buyer moves from point A on the demand curve DD at price OP to point B on the new demand curve DD, purchasing the same quantity (Oq) despite the price remaining unchanged. This increase in demand happens not because the price decreased but due to changes in factors like consumer income or other demand determinants. This is called an **Increase in Demand**.



Change in Quantity Demanded: Change in quantity demanded means a change in the quantity demanded due to a price change, while all other factors remain unchanged.

In the diagram, when the price decreases from P to P_1 , the quantity demanded increases from Q to Q_1 . Conversely, when the price increases from P_1 to P , the quantity demanded decreases from Q_1 to Q . The change in quantity demanded occurs along the same demand curve DD , moving from point E to point E_1 , and from point B to point C .

- ✓ **Change in Demand** is influenced by changes in external factors and causes a shift in the demand curve.
- ✓ **Change in Quantity demanded** results from price changes and is reflected as movement along the same demand curve.



Understanding these changes helps in analyzing the dynamics of demand and supply in the market.

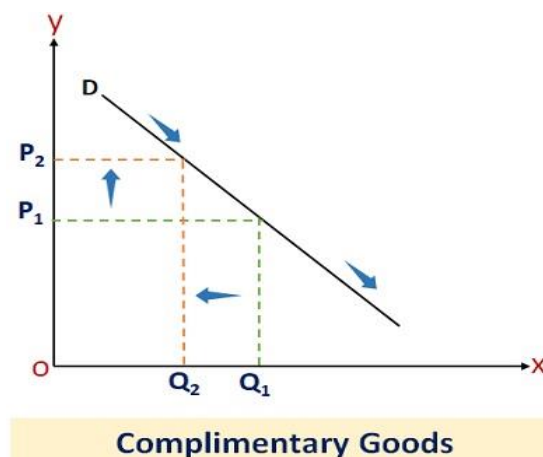
8. If the price of milk increase, what will happen to the demand for coffee? Explain with the help of graph. 97(3.c)

Effect of Increase in Milk Price on Coffee Demand

Milk and coffee are generally considered **complementary goods**, meaning they are often used together. When the price of milk increases, it can affect the demand for coffee, as many people use milk to prepare coffee.

Possible Impacts:

1. **Decrease in Demand:** When the price of milk rises, the cost of making coffee increases, which may reduce the demand for coffee. Consumers may become less interested in drinking coffee due to the higher cost of milk.
2. **Substitute Products in the Market:** Instead of increasing the price of coffee, consumers might turn to alternative beverages like tea or other sweet drinks that do not require milk.



When the price of milk increases from P_1 to P_2 , the demand for coffee decreases. As a result, the quantity demanded for coffee decreases from Q_1 to Q_2 . This indicates that milk and coffee act as complementary goods, where the price increase of one affects the demand for the other.

Market Equilibrium

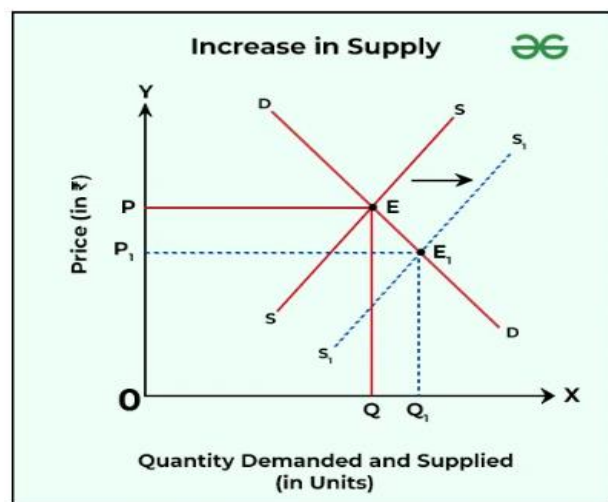
Market equilibrium is the position where the demand and supply of a product are equal. In this state, there is neither a shortage nor surplus of the product, and the price remains stable.

Example: Suppose the demand for a product is 100 units and the supply is also 100 units, with a price of \$10. If the price rises to \$12, the demand decreases, and supply increases, creating a surplus of the product in the market.

Impact of Subsidy on Equilibrium Price:

When the government provides a subsidy to control the price of a product, the following impacts may occur:

1. **Price Reduction:** Due to the subsidy, the price of the product decreases, which is beneficial for consumers.
2. **Increase in Demand:** As the price decreases, consumers are more willing to buy larger quantities of the product.
3. **Decrease in Supply:** Producers may have to sell at a lower price due to the subsidy, which reduces their profit and decreases supply.
4. **Market Imbalance:** Although demand increases, if supply decreases, there may be an imbalance in the market, leading to long-term issues.



Graph Explanation: If supply increases, the supply curve shifts from SS to S_1S_1 . The initial equilibrium point was E , with price P and quantity Q . After the supply increases, the new equilibrium point becomes E_1 , where the same price results in a higher quantity (Q_1) being supplied.

Impact on Equilibrium When Supply Changes:

When the supply of a product changes, it will affect the equilibrium price and quantity. Here's how:

1. **Increase in Supply:**

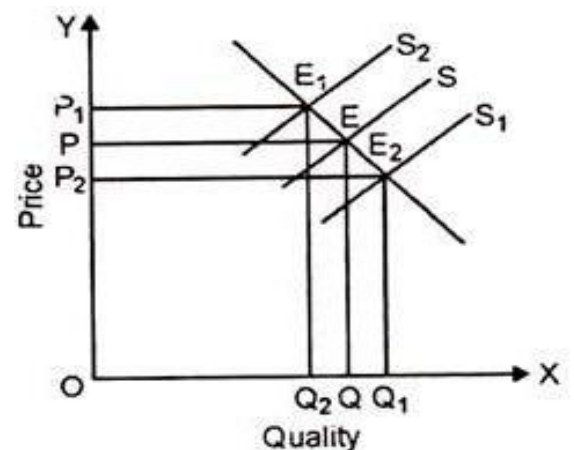
- ✓ **Effect on Price:** If supply increases (e.g., due to technological advancements or lower production costs), the supply curve shifts to the right. This leads to an excess supply at the original price, pushing the price down.
- ✓ **Effect on Quantity:** As the price decreases, more consumers are willing to buy the product, and the quantity demanded increases. Therefore, the equilibrium quantity rises.
- ✓ **Result:** The equilibrium price falls, and the equilibrium quantity increases.

2. Decrease in Supply:

- ✓ **Effect on Price:** If supply decreases (e.g., due to higher production costs or reduced availability of resources), the supply curve shifts to the left. This leads to an excess demand at the original price, pushing the price up.
- ✓ **Effect on Quantity:** As the price increases, fewer consumers are willing to buy the product, and the quantity demanded decreases. Therefore, the equilibrium quantity falls.
- ✓ **Result:** The equilibrium price rises, and the equilibrium quantity decreases.

Diagram Representation:

- ✓ If the supply increases, the supply curve shifts to the right ($S \rightarrow S_1$), leading to a lower price and higher quantity at the new equilibrium.
- ✓ If the supply decreases, the supply curve shifts to the left ($S \rightarrow S_2$), leading to a higher price and lower quantity at the new equilibrium.



In both cases, the shift in the supply curve leads to a new equilibrium price and quantity, depending on whether the supply has increased or decreased.

11. In Ramadan the demand of dates increase. What will be the impact on equilibrium in price and quantity?

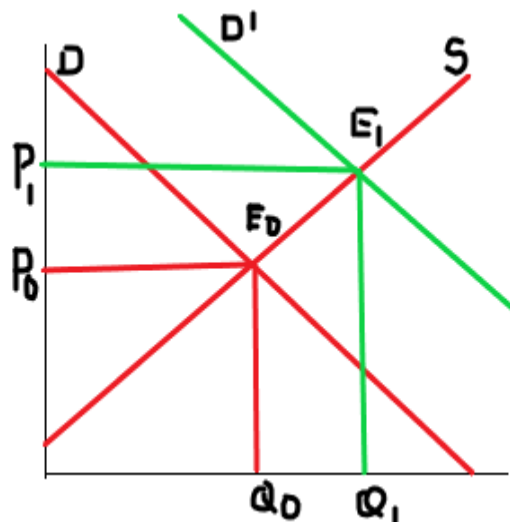
Impact of Increased Demand for Dates during Ramadan:

During the month of Ramadan, Muslims fast, and consuming dates becomes a common practice. This increases the demand for dates, which affects the market as follows:

1. **Increase in Demand:** During Ramadan, as the demand for dates increases, the demand curve shifts to the right. This results in a new demand position for the equilibrium price.
2. **Increase in Equilibrium Price:** As demand increases, consumers are willing to pay a higher price for dates, which helps to increase the market equilibrium price.
3. **Increase in Quantity:** Due to the price increase, producers will be motivated to produce more dates, leading to an increase in the quantity of dates in the market.
4. **Establishment of New Equilibrium:** A new equilibrium position is formed where both the price and quantity of dates increase.

Explanation with Graph:

- The demand curve shifts from D to D_1 due to the increased demand for dates during Ramadan.
- As a result, the equilibrium point moves from E_0 to E_1 .
- **Increase in Demand:** The increased demand leads to the creation of a new demand curve D_1 , indicating higher demand in the market.
- **Change in Equilibrium Point:** This shift results in the equilibrium point moving from E_0 to E_1 , where new prices and quantities are determined.
- **Price Increase Effect:** At the new equilibrium, the price of dates increases from P_0 to P_1 , and the quantity increases from Q_0 to Q_1 .



Thus, the increase in demand for dates during Ramadan has a significant effect on both the price and quantity in the market.

12. In the natural monopoly how price ceiling could be beneficial for common people? 99(5.b)

Natural Monopoly:

A natural monopoly occurs when a single firm operates in the market due to the lack of competition, or when multiple firms would lead to higher costs. Examples of natural monopolies include electricity, gas, and water supply systems.

Advantages of Price Ceiling in Natural Monopoly:

In natural monopoly markets, the government needs to set a price ceiling to ensure that the general public can access goods or services at a fair price. The benefits of setting a price ceiling are:

1. **Protection for Consumers:** Monopoly companies may set high prices, which can be burdensome for low-income individuals. By setting a price ceiling, consumers can access services at affordable rates.
2. **Reduction in Excess Profit:** Price ceilings can reduce the abnormal profits made by monopoly firms, benefitting the public by keeping prices fair.
3. **Market Stability:** Government-set price ceilings help maintain price stability in the market, preventing sudden price hikes.

4. **Fairness and Accessibility:** Price ceilings ensure that even low-income and middle-class people have easy access to essential services like electricity, gas, or water.

Example: In Bangladesh, the government determines the prices for electricity and gas to make these services affordable for everyone.

13. What is the price ceiling and price floor? 97(6.a)

Price Ceiling:

A price ceiling is the maximum price set by the government or authorities, above which a product or service cannot be sold legally. The purpose of a price ceiling is to keep a product or service affordable for consumers. An example of a price ceiling is rent control, where the government limits how much landlords can charge for rent.

Price Floor:

A price floor is the minimum price set by the government, below which a product or service cannot be sold legally. The purpose of a price floor is to protect the income of producers. An example of a price floor is the minimum price set for agricultural products, ensuring that farmers receive a guaranteed minimum price for their goods.

14. Explain the concept of Indifference Curve and Indifference Map. 98(3.a)

Indifference Curve:

An indifference curve is a graphical representation that shows different combinations of two goods that provide the same level of satisfaction or utility to a consumer. It indicates that if a consumer reduces the quantity of one good and increases the quantity of another, their overall satisfaction remains unchanged.

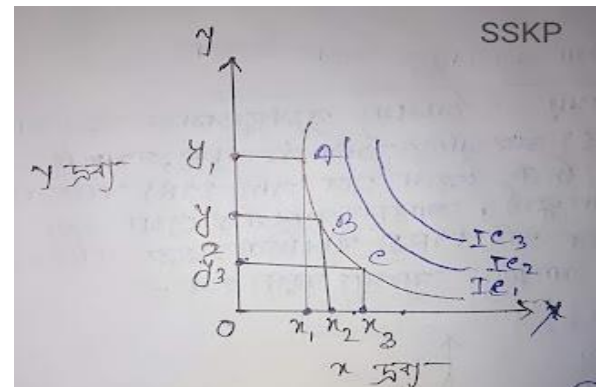
Characteristics of Indifference Curves:

1. **Downward Sloping:** Indifference curves are generally downward sloping, which means that to increase the quantity of one good, the quantity of the other good must be reduced.
2. **Infinite Number of Curves:** There are an infinite number of indifference curves because there are numerous combinations of goods that give the consumer different levels of satisfaction.
3. **Non-intersecting:** Two indifference curves cannot intersect because each curve represents a different level of satisfaction. A consumer cannot be at two different levels of satisfaction at the same time.

4. **Negative Slope:** The slope of an indifference curve, called the Marginal Rate of Substitution (MRS), shows how much of one good can be substituted for another while maintaining the same level of satisfaction.

Explanation with Diagram:

In the diagram, X and Y are two goods, and the various combination points A, B, and C provide the consumer with equal satisfaction. The path formed by these points is the indifference curve IC1. As the quantities of these two goods change, the consumer's new indifference curves are IC2, IC3, and so on. These indifference curves collectively form what is called the indifference map.



Indifference Map:

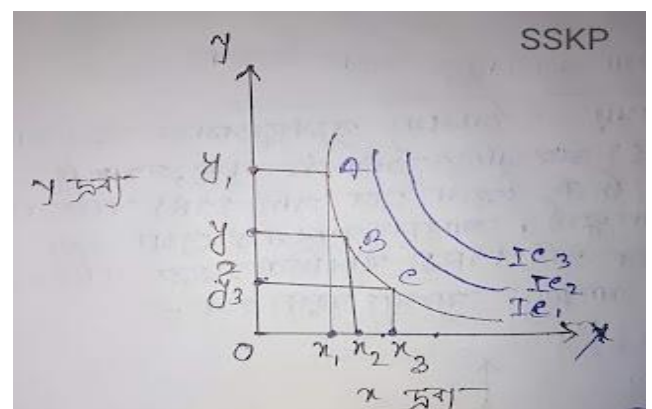
An indifference map is a graphical representation formed by multiple indifference curves. It shows the different combinations of goods that provide varying levels of satisfaction to the consumer.

Characteristics:

1. **Different Levels of Satisfaction:** Each indifference curve represents a specific level of satisfaction. Higher indifference curves indicate a higher level of satisfaction.
2. **Easy Analysis:** It provides a simple method for comparing consumer preferences, as it helps in analyzing the relationship between multiple goods and their combinations.
3. **Market Preferences:** An indifference map illustrates how consumers adjust their preferences to change the quantity of products and which combination provides the highest satisfaction.

Explanation with Diagram:

In the diagram, I1, I2, and I3 are three indifference curves. The combined form of these curves is the indifference map. The I3 curve represents the highest level of satisfaction, while the I2 curve indicates a lower level of satisfaction compared to I3. Similarly, the I1 curve shows a lower level of satisfaction than I2. Thus, when multiple indifference curves are drawn in a diagram, it is called an indifference map.



Indifference curves and indifference maps are important for understanding the relationship between consumer preferences and satisfaction levels. They are powerful tools for economic modeling and decision-making.

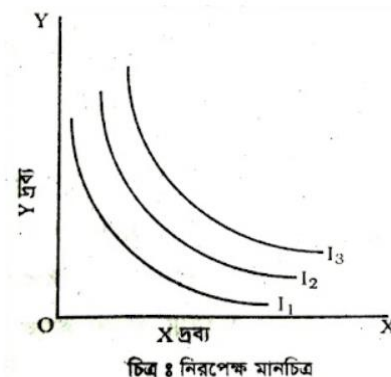
Indifference Curve:

An indifference curve represents the set of combinations of two goods that provide the same level of satisfaction to the consumer. Since the consumer derives the same level of satisfaction from all these combinations, they are indifferent between them. Therefore, this path of combinations is called the indifference curve. Below is an explanation with a diagram:

In the diagram, different combinations of goods X and Y, such as points A, B, and C, provide equal utility to the consumer. The path through these points is the indifference curve IC1. When the quantities of these two goods change, the consumer's new indifference curves, such as IC2, IC3, and so on, are formed. These indifference curves together form the indifference map.

Characteristics of Indifference Curves:

1. **Negative Slope:** Indifference curves typically have a negative slope, indicating that to increase the quantity of one good, the quantity of the other must decrease to maintain the same level of satisfaction.
2. **Convex to the Origin:** Indifference curves are convex to the origin, showing diminishing marginal rates of substitution.
3. **No Intersection:** Indifference curves do not intersect with each other, as this would imply two different levels of satisfaction for the same combination, which is impossible.
4. **Higher Indifference Curves Represent Higher Utility:** Higher indifference curves indicate higher levels of utility, as they represent combinations of goods that provide greater satisfaction.



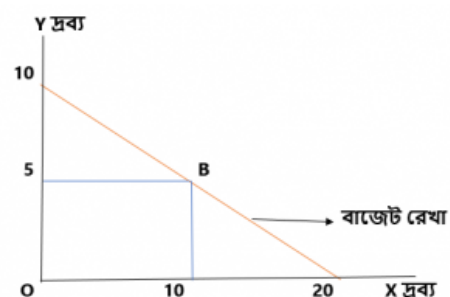
Thus, indifference curves are crucial in understanding consumer preferences and satisfaction levels.

Budget Line:

A budget line is a line that represents the combinations of two goods that a consumer can purchase with a given budget. Each point on the line indicates a different combination of goods that the consumer can buy, and the cost of each combination is equal to the total budget. In other words, a budget line shows the different combinations of two goods that a consumer can afford with their available income.

According to the diagram below, the budget line is plotted:

- The horizontal axis represents the quantity of good X, and the vertical axis represents the quantity of good Y.
- AC is the consumer's budget line. If the consumer spends all of their budget on good X, they can purchase OC (20 units) of good X.
- If the consumer spends all their budget on good Y, they can purchase OA (10 units) of good Y.
- If they decide to buy 10 units of good X, they can purchase 5 units of good Y, which is represented by point B.



The line formed by joining points A, B, and C is the budget line. Each point on this line corresponds to an equal total expenditure (100), so this is also called the Iso-cost line.

This budget line represents the trade-offs between the two goods that the consumer can make given their budget constraint.

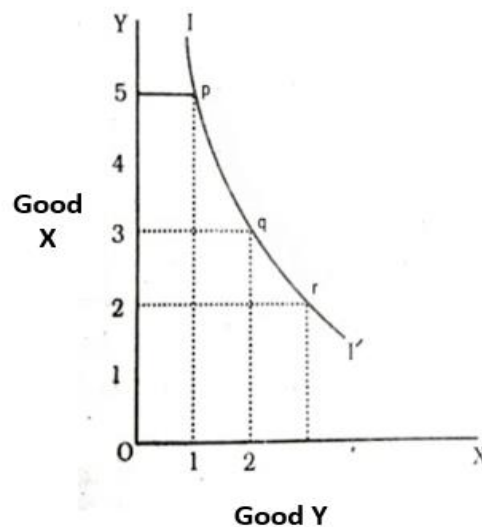
16. Why the Indifference curve is convex to the origin? Why two Indifference curves cannot intersect? Explain with the help of graph. 97(2.c)

Indifference Curve is Convex to the Origin:

Analysis from the Diagram:

In the above diagram, II' represents an indifference curve. According to point P on the indifference curve II', the consumer consumes 1 unit of good X and 5 units of good Y. Next, at point Q, to increase the consumption of good X by 1 unit, the consumer needs to reduce the consumption of good Y by 2 units. At point R, to increase the consumption of good X by an additional unit, the consumer must decrease the consumption of good Y by another unit.

Thus, we can conclude that the marginal rate of substitution (MRS) between good X and good Y is decreasing. As a result, the indifference curve II' is convex to the origin. The tendency of the indifference curve to be convex to the origin implies that the marginal rate of substitution (MRS) between good X and good Y is negative, meaning as the consumer consumes more of good X, they are willing to give up less of good Y.



Indifference Curves Never Intersect Each Other:

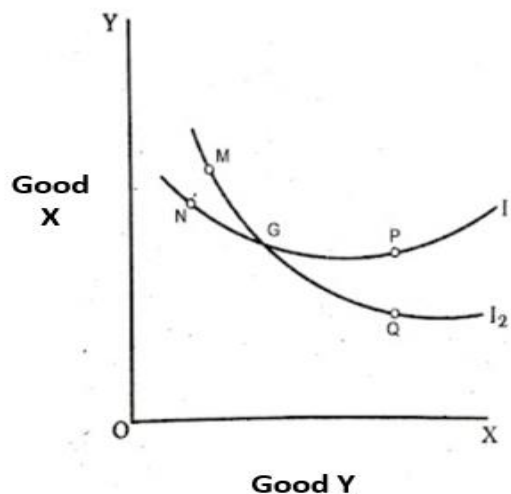
Analysis from the Diagram:

From the diagram, we can observe that I_1 and I_2 are two indifference curves that intersect each other. On the I_1 indifference curve, points M, G, and Q all lie on the same curve, indicating that the utility level at points M, G, and Q is equal. Similarly, on the I_2 indifference curve, points N, G, and P lie on the same curve, indicating equal utility at points N, G, and P.

We know that any point on the same indifference curve represents the same level of utility. So, considering the utility, we have $M = G$ and $N = G$, which should imply that $M = N$. However, this is not possible. Point M lies on a higher indifference curve than point N, meaning that the utility at point M is greater than at point N. Hence, $M > N$.

Similarly, as $G = Q$ and $G = P$, we should expect $Q = P$, but this is not the case based on the diagram. When we compare points Q and P, we see that $P > Q$ based on the utility level.

Therefore, if indifference curves were to intersect, it would imply that points M and N have the same utility at the intersection point, which contradicts the utility levels at other points. This inconsistency goes against the fundamental principle of indifference curves, which cannot intersect, as each curve represents a distinct level of utility.



17. Why the indifference curves are downward sloping?

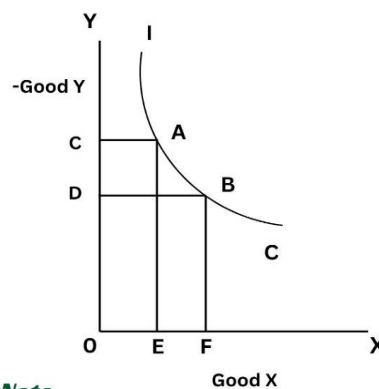
Indifference Curve Slopes Downwards to the Right:

One important characteristic of an indifference curve is that it slopes downwards to the right.

Significance: Every point on an indifference curve represents the same level of satisfaction or utility. However, these points indicate different combinations of two goods. Therefore, to maintain the same level of utility, if a consumer increases the quantity of one good, they must reduce the quantity of the other good. This relationship can only be explained by a rightward downward-sloping indifference curve.

Explanation with a Graph: In the graph, when the quantity of one good increases, the quantity of the other good decreases, but the level of utility remains unchanged.

Indifference curve downward sloping to the right



iEduNote

18. Total utility increasing at a decreasing rate-Explain with graph. 98(3.b)

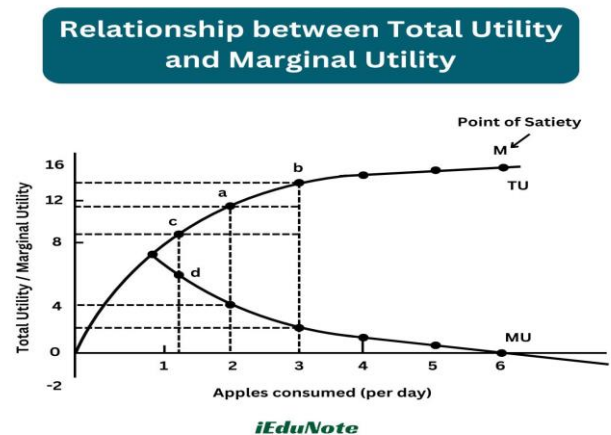
Diminishing Marginal Utility of Total Utility:

Total utility refers to the overall satisfaction or benefit a consumer gains from consuming a certain quantity of a good. When a consumer increases the quantity of a good, total utility generally increases. However, the increase in total utility often happens at a diminishing rate, meaning each additional unit of the good may add to the consumer's satisfaction, but the increase in satisfaction becomes smaller over time.

Total Utility and Marginal Utility Relationship:

Explanation with a Graph:

- **Total Utility Increase:** In the graph, total utility (Y-axis) increases as the quantity of a specific good (X-axis) increases. However, this is a gradually increasing curve.
- **Diminishing Increase:** As more units of the product are consumed, the increase in total utility becomes smaller. Initially, the first few units of the good provide a higher level of satisfaction, but as the consumer continues to increase the quantity, the satisfaction from each additional unit (marginal utility) decreases.



Marginal Utility: The change in utility that results from consuming one additional unit of a good is known as marginal utility. As total utility increases, marginal utility diminishes. This means that consumers experience higher satisfaction from the first few units of the product, but this satisfaction decreases with each additional unit.

The diminishing marginal utility of total utility suggests that while the first few units of a product may provide significant satisfaction, the additional units consumed result in less and less satisfaction. This concept is essential for understanding consumer behavior and demand for products.

19. Discuss the relationship between Total Utility (TU) and Marginal Utility (MU). 98(3.c)

Explanation of Total Utility and Marginal Utility:

At any given time, a consumer derives a certain amount of utility from consuming various units of a specific good. The total utility is the sum of the utilities derived from all the units of the good consumed.

For example, if a consumer consumes 3 units of good X, say X1, X2, and X3, the total utility derived from good X would be:

Total Utility (TU) for X =

$$TU_x = (\text{Utility from } X_1) + (\text{Utility from } X_2) + (\text{Utility from } X_3)$$

Marginal Utility (MU):

Marginal utility refers to the additional satisfaction or utility a consumer receives from consuming one more unit of a good, while keeping all other conditions constant.

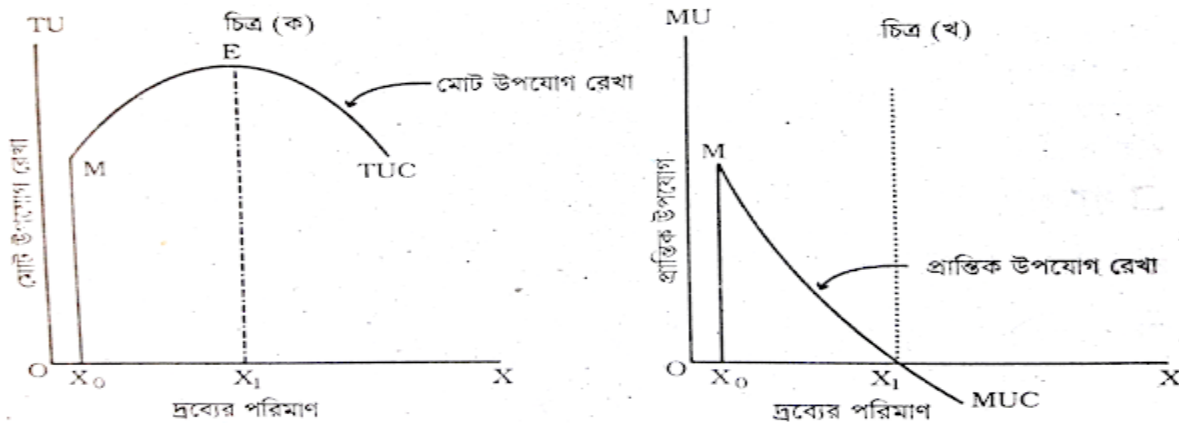
In symbolic form, marginal utility (MU) is calculated as:

$$MU = [TU \text{ for } n \text{ units}] - [TU \text{ for } (n-1) \text{ units}]$$

Where:

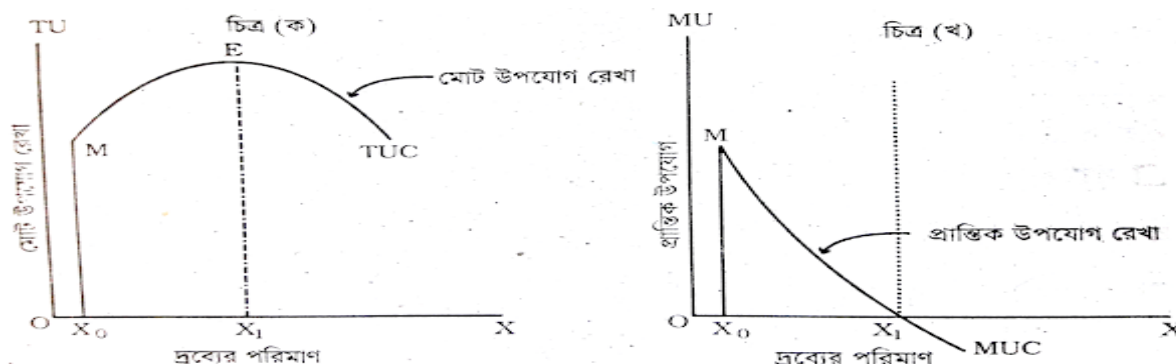
- **MU** = Marginal Utility
- **TU** = Total Utility
- **n** = Number of units consumed

Thus, marginal utility is the extra utility gained from the consumption of an additional unit of the good.



The relationship between total utility and marginal utility is described with the help of a diagram:

(a) In the above diagram, the horizontal axis shows the quantity consumed, and the vertical axis shows the total utility (TU). For OX0 units of consumption, the total utility is X0M. As the number of units of the good increases, the total utility increases, but at a diminishing rate. Therefore, the total utility curve (TUC) bends upward to the right. For OX1 units of consumption, the total utility is maximum (X1 E units). After OX1, if consumption is increased further, the total utility starts to decrease. Hence, after point E, the TUC slopes downward.



(b) In the diagram: The horizontal axis shows the quantity consumed, and the vertical axis shows the marginal utility. For OX_0 units of consumption, the total utility and marginal utility are equal (X_0M units). The marginal utility curve is derived from the slope of the total utility curve. From X_0 to X_1 , as consumption increases, total utility increases at a diminishing rate. As a result, the marginal utility curve slopes downward to the right. When the total utility is at its maximum for OX_1 units of consumption, marginal utility is 0 (zero). After OX_1 , if consumption increases further, total utility starts to decrease, and therefore, marginal utility becomes negative. Hence, after point X_1 , the marginal utility curve moves below the horizontal axis.

In light of the above discussion, the relationship between total utility and marginal utility is explained below:

1. Total utility is the sum of marginal utility, i.e., $TU = \sum MU$.
2. The change in total utility (ΔTU) divided by the change in quantity (ΔQ) gives marginal utility, i.e., $MU = \Delta TU \div \Delta Q$.
3. When total utility is increasing at an increasing rate, marginal utility starts to decrease.
4. When total utility reaches its maximum, marginal utility is zero.
5. When total utility starts to decrease, marginal utility becomes negative.

Thus, there is a deep relationship between total utility and marginal utility.

20. How would a demand curve look where the good is highly price elastic? Briefly explain. 99(3.c)সংক্ষেপে

Highly Price-elastic Goods:

Goods whose demand changes significantly in response to price changes are called highly price-elastic goods.

Demand Curve Shape:

For highly price-elastic goods, the demand curve is relatively flat and straight. This means that even a slight change in price leads to a large change in demand.

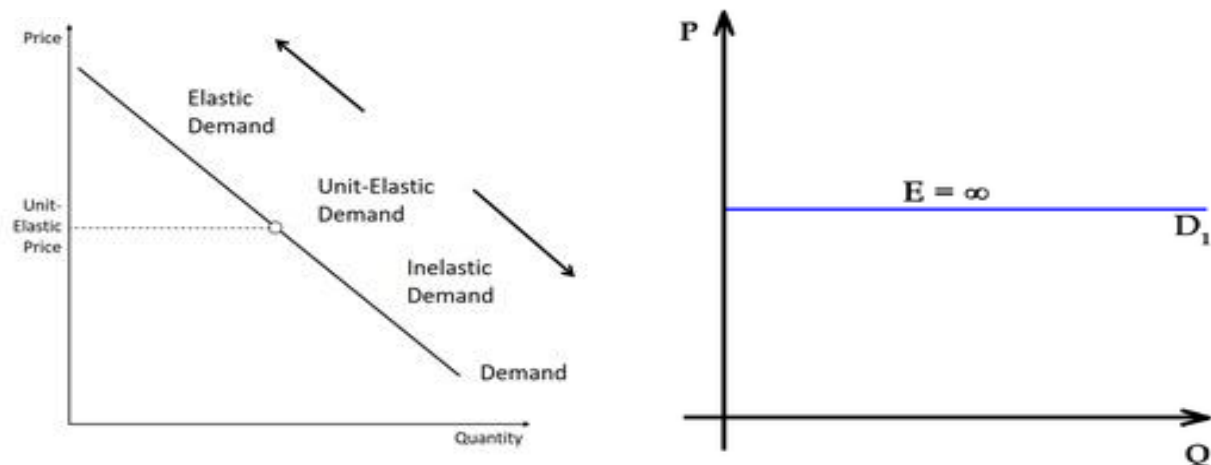


Diagram 1: Demand Curve for Normal Price-elastic Goods

Diagram 2: Demand Curve for Highly Price-elastic Goods

For highly price-elastic goods, the demand curve is flat and straight (due to significant demand changes).

Explanation:

Since for highly price-elastic goods, when the price increases, consumers may switch to substitute goods, their demand decreases significantly. Conversely, when the price decreases, demand increases significantly.

21. Draw a diagram of perfectly inelastic supply curve. 96(5.b)

Perfectly Inelastic Supply Curve

Definition: A perfectly inelastic supply curve refers to a situation where the quantity supplied remains unchanged despite changes in price. In other words, producers continue to supply the same amount of a product regardless of price increases or decreases.

Example:

Suppose the price of a product changes as follows:

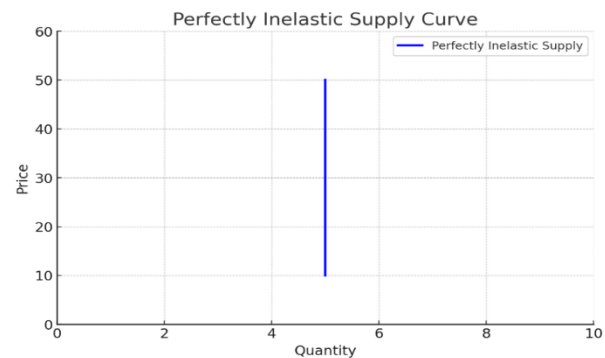
- Price 10 Taka: 5 units
- Price 20 Taka: 5 units
- Price 30 Taka: 5 units
- Price 40 Taka: 5 units
- Price 50 Taka: 5 units

Here, we can see that although the price has increased from 10 Taka to 50 Taka, the quantity supplied of the product remains constant at 5 units.

Chart:

(Chart would typically show a vertical supply curve indicating that supply is fixed at 5 units, regardless of price changes.)

Price(Taka)	Quantity Supplied(Unit)
10	5
20	5
30	5
40	5
50	5



22. Explain the four determinants of price elasticity of demand. 97(4.a)

Determinants of Price Elasticity

Price elasticity refers to how the demand or supply of a product responds to changes in price. There are several determinants that influence price elasticity, which are described below in four points:

1. **Necessity vs Luxury of the Product:** If a product is a necessity, its demand is less likely to decrease when the price increases, meaning the elasticity is lower. For example, the demand for food or medicine doesn't decrease much even if prices rise. On the other hand, if the product is a luxury, demand can quickly decrease with a price increase, like in the case of luxury cars or travel tickets.
2. **Availability of Substitutes:** If there are easily available substitute products for a good, its demand may decrease significantly when the price increases. For example, if the price of coffee rises, consumers may turn to tea or other beverages, increasing the price elasticity of coffee.
3. **Geographic Scope of the Product:** The geographic scope of the product affects price elasticity. Local products are generally less elastic because alternatives are harder to find. However, international products tend to show more elasticity due to the availability of more substitutes.
4. **Time Effect:** How quickly demand changes due to price changes depends on time. In the short term, price elasticity may be low because it is difficult for consumers to find alternative products quickly. However, in the long term, as consumers find substitutes, elasticity tends to increase.

These four determinants play a critical role in determining price elasticity, as they influence how consumers and markets respond to price changes. Understanding these factors helps businesses set prices and develop marketing strategies.

Income Elasticity of Demand

Income Elasticity of Demand refers to how the demand for a product changes when consumers' income changes. It measures the percentage change in demand resulting from a percentage change in income.

Formula for Income Elasticity of Demand

The formula used to calculate income elasticity is:

$$E_Y = \frac{\% \text{ change in quantity demanded}}{\% \text{ change in income}}$$

Where:

- E_Y is the income elasticity of demand.
- The numerator is the percentage change in demand, and the denominator is the percentage change in income.

Luxury Goods Demand Elasticity

Luxury goods are those products for which demand increases as people's income rises. When consumers' income increases, they tend to lean towards purchasing luxury goods, as these products usually provide additional benefits or exclusivity.

Why is Demand for Luxury Goods Elastic?

1. **Income Effect:** The income elasticity for luxury goods is usually greater than 1 ($E > 1$). For example, if consumers' income increases by 10%, the demand for luxury goods could increase by 20%.
2. **Lack of Substitutes:** Luxury goods often provide exclusivity or special features, leading consumers to be willing to purchase them as their income rises. Examples include foreign luxury cars or designer clothing.
3. **Social Recognition:** The use of luxury goods is often seen as a symbol of social status and recognition. As people become economically successful, they are more attracted to these types of products.

Income elasticity of demand helps to understand consumer behavior, especially when their income changes. Demand for luxury goods generally exhibits high elasticity, as people are more inclined to buy such items when their income rises, improving their standard of living.

Module-C: Production and Cost

1. What are the four factors of production? Discuss. 99(1.b)

The Four Factors of Production

1. Land:

- ✓ Land refers not only to physical land but also includes all natural resources such as water, minerals, forests, oil, etc. Land is crucial for production because it provides the essential resources required for producing all goods and services.
- ✓ **Example:** In Bangladesh's agricultural production, the importance of land is immense, as farmers need good land to increase crop yield.

2. Labor:

- ✓ Labor refers to the physical and mental effort of humans. The quality, skill, and experience of labor determine the quality of the produced goods.
- ✓ **Example:** In Bangladesh's various industrial factories, the skill and work quality of laborers impact the standard of production.

3. Capital:

- ✓ Capital refers to the machinery, equipment, infrastructure, and financial resources used for production. Capital helps make the production process faster and more efficient.
- ✓ **Example:** In Bangladesh's garment industry, the use of modern machinery has led to an increase in production.

4. Entrepreneurship:

- ✓ Entrepreneurship refers to the individual who coordinates the factors of production and brings new business ideas. The leadership and risk-taking ability of entrepreneurs enhance production efficiency.
- ✓ **Example:** Small and medium entrepreneurs in Bangladesh play a vital role in the economy by creating new products and entering new markets.

2. What is the law of diminishing marginal return? Discuss it with diagram. 96(2.a)

Limitations of the Law of Diminishing Marginal Utility

The **Law of Diminishing Marginal Utility** is an important concept in economics, stating that as an individual consumes more of a good or service, the additional utility (satisfaction) gained from each additional unit decreases. However, this law has several limitations, which are discussed below:

1. **Irreversible Conditions:** In some cases, utility may not diminish. For example, if a person is extremely thirsty, the utility gained from the first glass of water may be significantly higher than the subsequent glasses. The first unit consumed may provide a greater benefit than the others.
2. **Variety and Change:** If an individual consumes a variety of goods, the law of diminishing marginal utility may not apply. For instance, if a person consumes different types of food, the utility derived from each new dish may increase rather than decrease.
3. **Mental State:** Changes in a person's mental state can also influence utility. For example, if a person has a special liking for a particular food, the diminishing marginal utility law may not apply while eating that food, as they may continue to derive high utility even after consuming multiple servings.
4. **Scarcity of Resources:** When resources are scarce, the utility calculation can be different. For example, if a person is physically ill and lacks necessary medicine, the utility derived from each dose of medicine might feel much greater compared to a situation where the medicine is easily available.
5. **Combination of Different Goods:** When different products are consumed together, diminishing marginal utility may not occur. For example, when eating food and drinking beverages together, the combination of both may influence the utility differently, preventing the diminishing effect.

These limitations raise questions about the real-world application of the Law of Diminishing Marginal Utility, suggesting that utility concepts may vary in different contexts and the law's effectiveness can be limited in certain situations.

3. Define the relationship between Total Cost (TC), Average Cost (AC) and Marginal Cost in economics.

Relationship between Total Cost, Average Cost, and Marginal Cost

In economics, Total Cost (TC), Average Cost (AC), and Marginal Cost (MC) are interrelated concepts that are important for understanding production and cost management. Understanding their relationship is crucial for economic decision-making.

Total Cost (TC): Total Cost refers to the overall cost incurred by a firm or an economy in the production of goods and services. It is determined by adding both fixed and variable costs.

Formula:

$$TC = FC + VC$$

Where:

- **FC** = Fixed Cost
- **VC** = Variable Cost

Average Cost (AC): Average Cost is the cost per unit of output, calculated by dividing total cost by the number of units produced.

Formula:

$$AC = \frac{TC}{Q}$$

Where:

- **Q** = Quantity of output produced

Average Cost helps a producer understand the average cost of producing one unit of a product.

Marginal Cost (MC): Marginal Cost refers to the additional cost incurred when producing one more unit of output.

Formula:

$$MC = \frac{\Delta TC}{\Delta Q}$$

Where:

- **ΔTC** = Change in Total Cost
- **ΔQ** = Change in Quantity produced

Relationship:

- ✓ **Average Cost and Total Cost Relationship:** Average Cost depends on the Total Cost. As Total Cost increases, Average Cost also increases. However, the change in Average Cost is influenced by the rate of change in Total Cost and the number of units produced.
- ✓ **Marginal Cost and Average Cost Relationship:**
 - When Marginal Cost is below Average Cost, Average Cost decreases.
 - When Marginal Cost is above Average Cost, Average Cost increases.
 - When Marginal Cost equals Average Cost, Average Cost reaches its minimum point.

Example:

Understanding the relationship between Total Cost, Average Cost, and Marginal Cost is essential for production and cost management. These concepts work together to help businesses make accurate decisions regarding production and pricing.

4. Explain the relationship among Marginal Cost (MC), Average Cost (AC) and Average Variable Cost (AVC) with diagram. 99(4.b)

Relationship between Marginal Cost (MC), Average Cost (AC), and Average Variable Cost (AVC)

To understand the relationship between Marginal Cost (MC), Average Cost (AC), and Average Variable Cost (AVC), an ideal graph is used. Here's an explanation:

1. Marginal Cost (MC):

- ✓ Marginal Cost refers to the additional cost incurred for producing one more unit of output.
- ✓ Initially, MC decreases, and then it gradually increases as production expands.

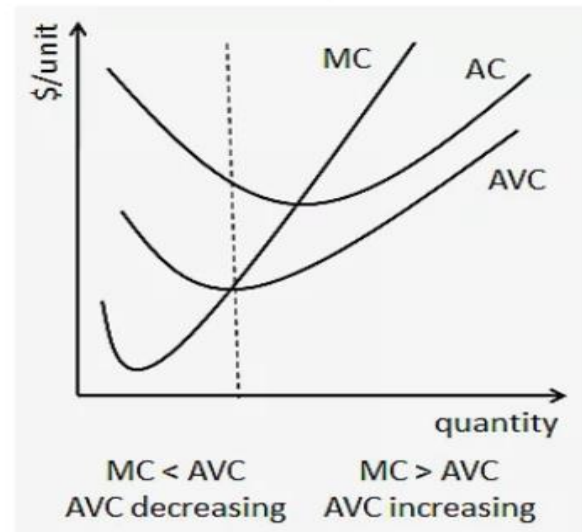
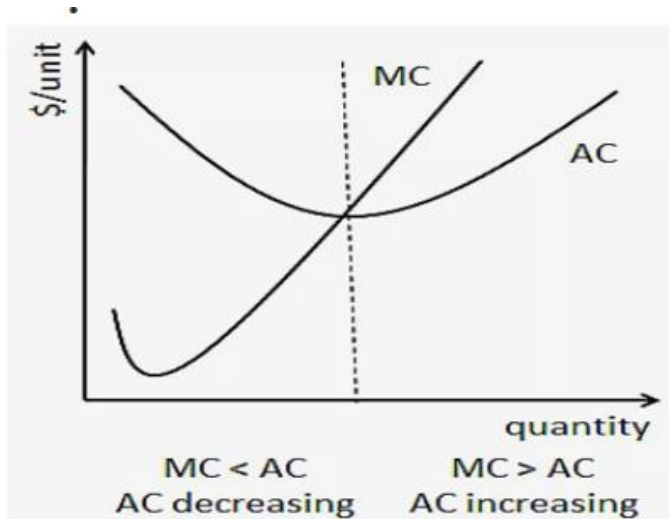
2. Average Cost (AC):

- ✓ Average Cost is the total cost per unit of output.
- ✓ Formula: **AC = Average Fixed Cost (AFC) + Average Variable Cost (AVC).**
- ✓ When MC is below AC, AC decreases; when MC rises above AC, AC increases.

3. Average Variable Cost (AVC):

- ✓ Average Variable Cost refers to the variable cost per unit of output.
- ✓ AVC typically follows a U-shaped curve and is always below the Average Cost (AC).
- ✓ When MC is below AVC, AVC decreases; when MC is above AVC, AVC increases.

This relationship is essential for understanding cost behavior and helps businesses make informed decisions regarding production and cost management.



In a typical cost curve graph:

1. **AC (Average Cost) vs. MC (Marginal Cost):**

- ✓ When **MC** is below **AC**, **AC** is **decreasing**.
- ✓ When **MC** is above **AC**, **AC** is **increasing**.
- ✓ When **MC** equals **AC**, **AC** is at its **minimum**.

2. **AVC (Average Variable Cost) vs. MC:**

- ✓ When **MC** is below **AVC**, **AVC** is **decreasing**.
- ✓ When **MC** is above **AVC**, **AVC** is **increasing**.
- ✓ When **MC** equals **AVC**, **AVC** is at its **minimum**.

This relationship forms the basis for understanding how costs behave in the short run and helps firms optimize their production levels.

Module-D: Market Structure

1. "The banking industry of Bangladesh follows monopolistic market structure"-Do you agree? Show your

Characteristics of a Monopoly Market:

In a monopoly market, there is typically a single dominant firm, entry barriers, and limited alternatives. However, the banking sector in Bangladesh is not fully monopolistic; rather, it exhibits oligopolistic characteristics.

Monopoly Features in Bangladesh's Banking Sector:

1. **Limited Competition:** In Bangladesh, state-owned banks (such as Sonali Bank, Janata Bank) control a large portion of the market, indicating some level of monopoly dominance.
2. **Entry Barriers:** Establishing a new bank requires approval from the central bank, which creates an entry barrier.
3. **Limited Autonomy in Interest Rate Determination:** Bangladesh Bank regulates interest rates, which reduces the freedom for competition in terms of setting rates.
4. **Influence of Large Banks:** Larger state-owned and private banks enjoy more benefits compared to smaller banks.

Why It Is Not Fully Monopolistic:

1. **Competitive Banking System:** Bangladesh has over 60 banks, including government, private, foreign, and Islamic banks, which creates competition in the market.
2. **Alternative Options for Customers:** Customers can avail services from various banks, reducing the potential for a single bank to dominate the market.
3. **Technological Competition:** The expansion of mobile banking and digital transactions has created opportunities for competition.

Therefore, while Bangladesh's banking sector shows some monopolistic features, it is not entirely a monopoly but rather an oligopoly, where a few large banks control the market. However, competition, technological advancement, and the entry of new banks prevent it from becoming a fully monopolistic market.

2. Assess the banking industry of Bangladesh based on the features of the market structure and comment on the market structure of the banking industry in Bangladesh. 97(5.c)

Evaluation of the Market Structure of Bangladesh's Banking Sector

The market structure of Bangladesh's banking sector is based on an oligopoly, with a few large banks and many smaller banks operating within the sector. The following points outline the characteristics of this market:

1. **Limited Number of Large Banks:** A few large banks, such as BRAC Bank, City Bank, and The HSBC Bank, control a significant portion of the market. These banks typically offer better services and benefits.
2. **Level of Competition:** There is competition among banks, but in some cases, they are interdependent on each other. As a result, there is a certain degree of coordination among banks in terms of pricing and setting interest rates.
3. **Product and Service Differentiation:** Banks offer a variety of products and services, such as savings accounts, loans, card services, and digital banking. This differentiation helps attract customers.
4. **Government Regulation:** Bangladesh Bank, the country's central bank, regulates the banking sector. It plays a crucial role in maintaining market stability and managing risks.
5. **Market Entry and Exit:** There are regulations and conditions for new banks to enter the banking sector, creating barriers to entry. As a result, it is difficult for new competitors to enter the market.

Opinion:

The market structure of Bangladesh's banking sector is characterized by oligopolistic features. A few large banks dominate the market and stay alert to each other's activities. Due to government regulation and certain restrictions, it is challenging for new entrants to break into the market, which limits competition.

Therefore, to make Bangladesh's banking sector more competitive, it is necessary to reduce entry barriers and create opportunities for new banks. This will ensure better services and fair pricing for consumers.

3. What is perfect competition? Discuss the underlying assumptions in a perfectly competitive market model.
99(3.b)

Perfectly Competitive Market:

A perfectly competitive market is one where many buyers and sellers exist, and the price of goods is determined by the market. No single seller or buyer can influence the price because there are many sellers offering the same type of product.

Assumptions of a Perfectly Competitive Market:

1. Large Number of Buyers and Sellers:

- ✓ There are many buyers and sellers in the market.
- ✓ No buyer or seller can influence the price because they buy or sell only a small portion of the total demand or supply.

2. Homogeneous Products:

- ✓ All sellers produce identical products.
- ✓ There is no qualitative difference between products from different sellers, and therefore, the price remains the same.

3. Perfect Knowledge of the Market:

- ✓ Both buyers and sellers have complete information about the market, prices, demand, and supply.
- ✓ There is no opportunity to charge extra prices.

4. Free Entry and Exit:

- ✓ New firms can enter the market, and existing firms can leave the market freely.

5. Complete Mobility of Factors of Production:

- ✓ Factors of production can move freely and independently across sectors.
- ✓ There are no social or political barriers.

6. Same Price for the Same Product:

- ✓ The price of the product is determined by supply and demand, and it is the same everywhere.
- ✓ No party can change the price.

7. Perfect Substitutes:

- ✓ Due to competition in the market, the price change of one product influences the price of other products.

8. **Absence of Bias:**

- ✓ The market is impartial, and no seller can sell at a lower price or a buyer can buy at a higher price.

9. **Rational Behavior:**

- ✓ Buyers behave rationally, choosing to buy quality products at the lowest prices.

These conditions illustrate the functionality and characteristics of a perfectly competitive market.

4. What is Perfect Competition? Discuss the Break Even Point. 96(4.b)

Perfectly Competitive Market:

A perfectly competitive market is a market structure where there are numerous buyers and sellers, and no individual or firm can control the market price. The main characteristics of this market are:

1. **Numerous Sellers and Buyers:** Each firm is a small producer, and they have no control over the market price.
2. **Free Entry and Exit for All:** New firms can easily enter the market, and existing firms can exit if necessary.
3. **Homogeneous Products:** All products in the market are identical and interchangeable.
4. **Perfect Information Availability:** Both buyers and sellers have complete information about the prices and quality of products.

Example: The markets for rice, salt, and wheat in Bangladesh are close to a perfectly competitive market in some cases.

Break-Even Point:

The break-even point is the point at which total revenue equals total cost. In other words, at this point, the business makes neither profit nor loss. To determine the break-even point:

- **Fixed Cost:** Costs that do not depend on the quantity of production (e.g., rent, salaries).
- **Variable Cost:** Costs associated with the level of production (e.g., raw materials).
- **Total Cost = Fixed Cost + Variable Cost.**

To determine the break-even point, set the total revenue equal to total cost in an equation.

Monopolistic Competition:

Monopolistic competition is a market structure where many sellers sell similar but slightly differentiated products. This typically occurs in markets with a limited variety of products, such as fashion, restaurants, or hotels. The characteristics of monopolistic competition are:

1. **Many Sellers:** There are many sellers in the market who offer similar but slightly different products.
2. **Product Differentiation:** The products offered are distinct from each other, providing customers with various alternatives.
3. **Limited Control Over Price:** Each seller has some control over the price of their product, but this control is limited.
4. **Free Entry and Exit:** New businesses can enter the market, and existing businesses can exit if they are no longer profitable.

Difference from Monopoly:

- **Number of Sellers:** In a monopoly, there is only one seller who controls the entire market, whereas in monopolistic competition, there are many sellers.
- **Price Determination:** In monopolistic competition, sellers have some power to set prices, but in a monopoly, the seller has complete control over the pricing.
- **Product Variety:** In a monopoly, a single product is sold, while in monopolistic competition, there is product differentiation among the goods offered.

Collusion:

Collusion is a situation where a number of firms make agreements with each other to control market prices, production, or profits through strategic actions. This is commonly seen in oligopoly markets.

Why is Maintaining a Cartel Difficult in an Oligopoly Market?

A cartel is a formal or informal agreement in which a few firms in an oligopoly market work together instead of competing. However, maintaining a cartel is difficult due to the following reasons:

1. **Incentive to Cheat:** Each firm may secretly break the agreement and sell more products to earn higher profits.
2. **Legal Barriers:** In many countries, cartels are illegal, as per laws like the Bangladesh Competition Act.
3. **New Entrants Cause Disruption:** When new firms enter the market, the cartel plans can be disrupted.
4. **Instability and Demand Changes:** If there is a decrease in demand, disagreements can arise between cartel members.
5. **Personal Gain:** Each firm seeks to increase its profit by exploiting the cartel's benefits, which can eventually make the cartel ineffective in the long run.

Example:

In the global oil market, OPEC (Organization of Petroleum Exporting Countries) is a cartel that attempts to stabilize prices by limiting oil production. However, the agreement often fails because member countries sometimes sell more oil than agreed, which disrupts the cartel.

7. Define oligopoly market. 97(5.a)

Oligopoly Market

Oligopoly is a market structure where a few large producers or sellers control the market. The main characteristics of this market are as follows:

1. **Limited Competition:** In an oligopoly, a few large companies dominate the market, and each company's decisions affect the actions of the other companies.
2. **Product Homogeneity or Differentiation:** Products in an oligopoly can either be homogeneous (e.g., steel or cement) or differentiated (e.g., cars or phones). When products are differentiated, companies compete through advertising and branding.
3. **Influence on Prices:** In an oligopoly, competitors typically watch each other's prices closely. If one company raises its price, other companies may respond by adjusting their prices as well.
4. **Cartel Formation:** Sometimes, oligopoly companies may form a cartel, where they agree on prices and control the market. This reduces competition and helps to increase prices in the market.

8. In the natural monopoly how price ceiling could be beneficial for common people? 99(5.b)

Natural Monopoly:

A natural monopoly occurs when a single firm operates in the market because there is no scope for competition or when multiple firms would increase costs.

Example: Electricity, gas, water supply systems, etc.

Advantages of Price Ceiling:

In a natural monopoly market, the government needs to set a price ceiling so that people can access goods or services at a fair price.

1. **Consumer Protection:** Monopoly companies can set high prices, which can be difficult for the poor. By setting a price ceiling, people can get services at affordable prices.
2. **Reduction of Excess Profit:** By reducing the abnormal profits of monopoly firms, the public can benefit.
3. **Market Stability:** Setting prices by the government keeps the market price stable and prevents sudden price hikes.
4. **Fairness and Access:** Poor and middle-income people can easily access electricity, gas, or water.

Example: In Bangladesh, the government sets the prices of electricity and gas to make them affordable for everyone.

9. In monopoly there is no supply curve-Explain. 96(4.c)

Reasons for the Absence of a Supply Curve in a Monopoly:

1. **Single Seller:** In a monopoly, the single seller controls the market price, and as a result, there is no fixed supply curve. Therefore, their supply level in the market is not defined.
2. **Price Determination:** In a monopoly, the seller has the power to raise prices. This keeps the supply curve unchanged because the seller fears a decrease in demand when prices are raised.
3. **Barriers to Entry:** When new competitors cannot enter the market, the seller maintains the power to set prices, which affects the supply curve.
4. **Centralized Control:** The seller in a monopoly has complete control over production, so their supply curve typically changes according to production levels and market demand.

For these reasons, a traditional supply curve does not exist in a monopoly, and it is entirely influenced by market demand.

10. Define Accounting profit and Economic profit. 96(6.c)

Accounting Profit:

Accounting profit is the profit obtained by subtracting all explicit costs from total revenue.

Example: If a company's total revenue is \$100,000 and explicit costs are \$70,000, then the accounting profit will be:

$$100,000 - 70,000 = 30,000$$

Economic Profit:

Economic profit is the profit obtained by subtracting all explicit costs and implicit costs from total revenue.

Example: If total revenue is \$100,000, explicit costs are \$70,000, and implicit costs are \$20,000, then the economic profit will be:

$$100,000 - (70,000 + 20,000) = 10,000$$

Module-E: Market Failure and Government Intervention

1. What is market failure? What are the key factors responsible for market failure? 97(6.b)

Market Failure:

Market failure occurs when the market is unable to allocate resources efficiently on its own, leading to a decrease in economic efficiency and overall welfare.

Main Causes of Market Failure:

1. **Externalities:** When the activities of a company affect others in a positive or negative way, and prices do not adjust accordingly.
2. **Lack of Information on Goods and Services:** When consumers or producers do not have enough information, they cannot make the right decisions, which leads to inefficient market management.
3. **Infrastructure Issues:** Market failure can also occur when competition is limited, such as in monopolies or oligopolies.
4. **Economic Power and Inequality:** When certain firms have too much power, they can set prices at will, reducing market efficiency.

2. What is public good? Define 'free rider problem' in context of public goods. 99(5.a)

Public Good:

A public good is a product or service that is essential, non-excludable, and non-rivalrous. This means:

- When one person uses it, it does not reduce the availability for others.
- No one can be excluded from using it.

Examples: National defense, street lighting, law enforcement, river dams, etc.

Free Rider Problem:

The free rider problem occurs when individuals enjoy the benefits of a public good but do not want to pay for it.

- **Reason:** Since public goods cannot exclude people from usage, many believe they will benefit without paying, relying on others' contributions.

- **Result:** If the government or institutions do not enforce mandatory taxation, it becomes difficult to ensure the adequate provision of public goods.

Example: If everyone believes the government will install street lights without paying taxes, tax collection will decrease, and the provision of public goods will be disrupted.

3. What is external benefit? How positive externalities and internalized using corrective subsidies? 98(6.b)

Positive Externality:

A positive externality occurs when an individual or organization provides additional benefits to others through their actions, but they do not receive any compensation for those benefits.

Examples:

1. **Education** – An educated person raises awareness in society, benefiting others as well.
2. **Vaccination programs** – When one person gets vaccinated, others are also protected.

How Corrective Subsidy Addresses Positive Externalities:

A corrective subsidy is a government policy where financial support is provided for activities that create positive externalities. This encourages individuals and organizations to engage in more beneficial actions.

How it works:

- **Encouraging individuals or organizations:** If education creates positive externalities, the government can provide subsidies to educational institutions, encouraging more people to pursue education.
- **Increasing social benefits:** By providing subsidies for vaccinations, more people will get vaccinated, making the entire society safer from diseases.
- **Ensuring the correct level of production:** Private organizations often do not account for the external benefits, so the government incentivizes them with subsidies to increase activities beneficial to society.

Examples:

- **Subsidies in education** → more educated people → skilled workforce → economic growth.
- **Subsidies for eco-friendly technology** → less pollution → improved quality of life.

4. Define positive externality and negative externality with example. 96(6.a)

Positive Externality:

A positive externality occurs when an individual or organization creates benefits for others. It has a positive impact on society, but the producer does not receive any economic compensation for it.

Examples:

1. **Education:** When an individual receives higher education, their skills improve, and it has a positive impact on society. This leads to improved work efficiency, better health, and a reduction in crime rates.
2. **Public Health Programs:** Vaccination programs improve people's health, benefiting society as a whole. The spread of diseases is reduced, and public health is enhanced.

Negative Externality:

A negative externality occurs when an individual or organization causes harm to others. This leads to negative effects on society.

Examples:

1. **Pollution:** When a factory causes air pollution, it harms the health of the local population. The risk of diseases increases due to pollution.
2. **Smoking:** Smoking can cause health problems for those nearby. Smokers not only harm themselves but also negatively affect others.

5. How does a proper tax system benefit the economy? 98(6.c)

An **efficient tax system** is crucial for the economy because it ensures development and fairness.

Benefits of an Efficient Tax System:

1. **Increase in Government Revenue:** The government can use tax collection to fund spending on education, healthcare, infrastructure, etc., which contributes to the country's development.
2. **Reduction in Socio-economic Inequality:** A progressive tax system (where higher-income individuals pay more taxes) reduces the gap between the rich and the poor, ensuring social justice.

3. **Increase in Investment and Growth:** If tax rates are fair and properly set, businesses thrive, and new investments flow in, leading to job creation and economic growth.
4. **Control of Inflation:** The government can use the tax system to control excess money flow, which helps reduce inflation.
5. **Environmental Protection:** Pollution taxes or carbon taxes encourage eco-friendly activities and help tackle climate change.
6. **Market Stability:** An efficient tax system helps maintain market stability without placing undue pressure on businesses.

Examples:

- In **Bangladesh**, income tax and Value Added Tax (VAT) help increase government revenue.
- In **India** and **Europe**, carbon taxes have boosted investment in environmentally friendly industries.

6. Discuss the pros and cons of subsidy? Do you think agricultural subsidy should be increased in our

Advantages of Subsidies:

1. **Increase in Production:** Subsidies allow farmers and industrialists to increase production at lower costs.
2. **Price Balance:** Subsidies help maintain lower prices in the market, which is beneficial for the general public.
3. **Employment Creation:** Subsidies in agriculture and industry sectors create new job opportunities.
4. **Increase in Exports:** Subsidies on agricultural or industrial products boost production and enhance export competitiveness.
5. **Food Security:** Subsidies in agriculture ensure adequate food supply and guarantee food security in the country.

Disadvantages of Subsidies:

1. **Budget Pressure:** The government incurs additional expenses, which may reduce spending in other sectors.
2. **Waste and Inefficiency:** Subsidies are often misused, leading to inefficiencies.
3. **Market Distortion:** With subsidies, some businesses may create artificial shortages.
4. **Tax Burden:** The public may face higher taxes as the government spends more on subsidies.

Should Subsidies in Agriculture be increased in Our Country?

It should be increased because:

- **Bangladesh is an agriculture-dependent country**, where the majority of people rely on agriculture.
- Subsidies on fertilizers, electricity, irrigation, and seeds would reduce the production costs for farmers.
- Increased food production will lower prices, benefiting the general population.
- It will be possible to increase agricultural exports.

Increasing subsidies in agriculture would boost production, ensure food security, and benefit farmers. However, proper management is essential to avoid wastage.

7. "Asymmetric information may cause moral hazard problems in the banking industry"-Explain. 97(6.c)

Asymmetric Information:

Asymmetric information occurs when one party possesses more or better information compared to the other party. In the banking sector, this typically occurs between the borrower and the lender (bank).

Moral Hazard:

Moral hazard is a situation where the behavior of one party can be harmful to the other party. In the banking sector, asymmetric information can create moral hazards for the following reasons:

1. **Borrower's Behavior:** When banks provide loans, borrowers often do not share accurate information about their financial condition. If borrowers believe that the bank is not monitoring their actions, they may take on more risk, which can lead to losses for the bank.
2. **Economic Crisis:** If borrowers adopt fraudulent practices to make a profit, the bank may face an economic crisis. They may fail to recover loans, which leads to a deterioration in the financial position of the bank.
3. **Bank's Reliability:** When banks lend money, they may lose reliability due to a lack of accurate information. If a borrower claims to run a profitable business but is not, the bank will incur financial losses.
4. **Market Instability:** Asymmetric information can lead to instability in the banking sector, where some banks may take on excessive risk, negatively affecting the entire banking system.

Asymmetric information in the banking sector can create moral hazards, damaging the relationship between lenders and borrowers. To address these issues, transparency, information exchange, and regulation are essential.

Module-F: Basic Macroeconomics

1. What is the policy objectives of Macro Economics? 96(9.a)

The Objectives of Macroeconomic Policy:

The main objective of macroeconomic policy is to ensure the economic stability, growth, and welfare of a country. The key objectives are as follows:

1. **Economic Growth:** The goal of macroeconomics is to increase the country's total production (GDP). This is achieved through industrialization, technological advancement, and improving labor efficiency.
2. **Control of Inflation:** Policies are implemented to keep the inflation rate stable, ensuring that the prices of goods do not rise excessively. High inflation damages people's purchasing power.
3. **Reducing Unemployment:** The aim of macroeconomics is to achieve full employment of the labor force in the country. By reducing unemployment, people's incomes increase and the standard of living improves.
4. **Balance of Trade:** Maintaining a balance between imports and exports of the country. This helps stabilize the currency and increase foreign exchange reserves.
5. **Economic Equality:** One of the goals of macroeconomics is to ensure economic equality, reducing income disparities between different levels of society.
6. **Increase in Social Welfare:** The government's aim is to provide social services and welfare programs for citizens, improving the standard of living and enhancing public welfare.

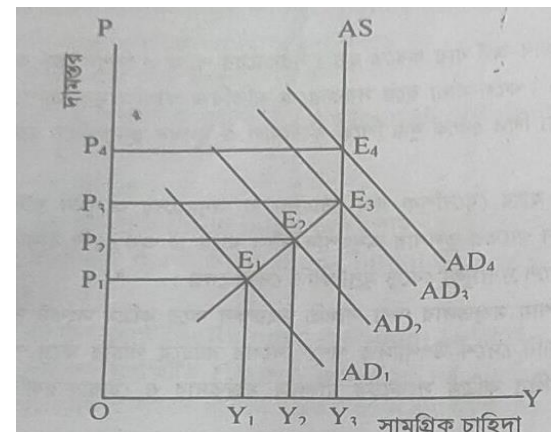
2. Define the demand pull inflation and cost push inflation with diagram. 96(9.b)

Demand-Pull Inflation:

The diagram on the left illustrates Demand-Pull Inflation. In this scenario, an increase in aggregate demand (AD) results in both higher price levels and increased output.

Diagram Analysis:

- Initially, the aggregate demand curve (AD_1) and the aggregate supply curve (AS) intersect at the equilibrium point E_1 , where the price level is P_1 and the output is Y_1 .



- As demand increases, the AD curve shifts from AD_1 to AD_2 , AD_3 , and finally to AD_4 .
- As a result, the price level increases from P_1 to P_2 , P_3 , and P_4 , and output increases from Y_1 to Y_2 , Y_3 .

Cause: Generally, excessive demand in the economy, increased government spending, tax cuts, expansionary monetary policies, and an increase in the money supply contribute to this inflationary pressure.

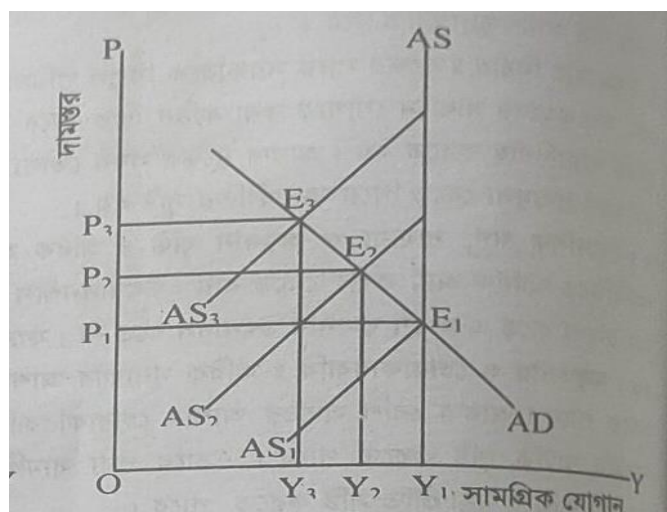
Cost-Push Inflation:

The diagram on the right illustrates Cost-Push Inflation. In this scenario, an increase in production costs leads to a contraction in supply, causing the price level to rise.

Diagram Analysis:

- Initially, the aggregate supply curve (AS_1) and the aggregate demand curve (AD) intersect at the equilibrium point E_1 , where the price level is P_1 and the output is Y_1 .
- As production costs rise, the aggregate supply curve shifts from AS_1 to AS_2 and then to AS_3 .
- As a result, the price level increases from P_1 to P_2 and P_3 , while output decreases from Y_1 to Y_2 and Y_3 .

Cause: The increase in wages, raw material prices, production costs, fuel prices, etc., contributes to cost-push inflation.



3. Is inflation always bad? Explain. 96(9.c)

Inflation is not always bad, but its impact depends on the situation. Here are some points to consider:

1. **Low Inflation:** If a country experiences stable and controlled low inflation (e.g., 2-3%), it can be healthy for the economy. It encourages consumers to spend more and stimulates business investment.
2. **Growth Supportive:** If inflation is related to overall economic growth, it can be a positive sign. When workers' incomes rise, they are able to spend more, which increases sales for businesses.
3. **Beneficial for Borrowers:** When inflation rises, borrowers benefit because they have taken loans at fixed interest rates and will repay with money that is worth less in the future.
4. **Dangerous Inflation:** However, high and unstable inflation (e.g., 10% or more) can be dangerous for the economy. It erodes purchasing power and creates lasting uncertainty. This

leads to reduced savings and negatively impacts investments.

5. **Negative Effects of Inflation:** High inflation can lead to a decline in the value of money, which can cause social instability. It creates inequality in social justice and reduces the standard of living for ordinary people.

4. A perfectly competitive firm earns zero or normal profit in the long run why and how? 98(5.b)

A perfectly competitive firm earns zero or normal profit in the long run mainly due to the entry of new firms and market competition.

Why does zero or normal profit occur?

1. **Free Entry and Exit:** If a firm earns excess profit in the short run, new firms will enter the market, increasing supply, which will cause the market price to decrease.
2. **Excess Profit in the Short Run:** Some firms may earn profit in the short run, but due to competition, this excess profit will eventually reduce to zero in the long run.
3. **Increased Competition in Profitable Markets:** When new firms enter, supply in the market increases and the price decreases, bringing profit back to normal levels.
4. **Difference between Economic Profit and Accounting Profit:** In the long run, only normal profit remains, which is enough to keep entrepreneurs investing.

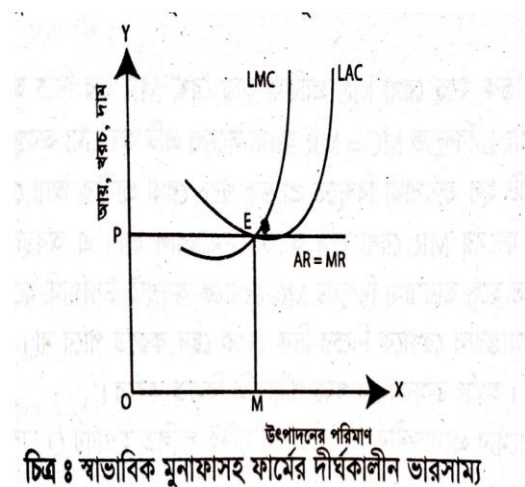
How does zero profit occur? (Explanation with a diagram)

In the diagram, the OX axis represents the quantity of production, and the OY axis represents price/cost/revenue. The AR = MR line indicates the average and marginal revenue of the industry. The LAC and LMC lines represent long-run average and marginal costs, respectively. At point E, the LMC line intersects the MR line from below. Hence, E is the equilibrium point. The equilibrium production is OM, and the price is OP.

In this case, $LMC = MR = AR = LAC = P$, as shown in the diagram.

- Total Revenue, $TR = OPEM$
- Total Cost, $TC = OPEM$
- Profit = $AR - TC = OPEM - OPEM = 0$

Therefore, in this condition, all firms earn normal profit.



1. **P = MC = ATC (Long-Run Equilibrium):** When price (P) is equal to average total cost (ATC), the firm earns zero economic profit.
2. **No New Firms Enter if No Extra Profit:** Hence, the market comes to equilibrium.
3. **If there is a Loss, Some Firms Exit:** This reduces supply, and the price returns to the normal level.

Thus, due to competition, no firm can earn excess profit in the long run, only normal profit.

5. Define and distinguish nominal interest rate and real interest rate. 97(9.a)

Nominal Interest Rate:

The nominal interest rate is the announced interest rate for loans or investments, which is determined without considering inflation or other economic changes. It directly functions as the market interest rate.

Real Interest Rate:

The real interest rate is calculated by subtracting the inflation rate from the nominal interest rate. It indicates the actual return on a loan or investment based on the real economic situation.

Differences:

1. Definition:

- ✓ **Nominal Interest Rate:** The announced interest rate for loans or investments.
- ✓ **Real Interest Rate:** The amount remaining after subtracting inflation from the nominal interest rate.

2. Calculation:

- ✓ **Nominal Interest Rate:** The direct market interest rate.
- ✓ **Real Interest Rate:** Nominal interest rate - inflation rate.

3. Economic Impact:

- ✓ **Nominal Interest Rate:** Indicates the attractiveness of investments, but the real interest rate reflects the actual profit of an investment based on economic realities.

6. What is employment? Explain the different types of unemployment in an economy. 97(7.a)

Employment:

Employment refers to an economic condition where individuals or workers are engaged in work and receiving compensation for their labor. Employment is not just the process of working but is also crucial for economic growth and social development.

Different Types of Unemployment:

1. **Voluntary Unemployment:** Occurs when workers choose not to work, such as waiting for a better job opportunity.
2. **Involuntary Unemployment:** Happens when workers lose their jobs against their will, such as due to an economic recession.
3. **Cyclical Unemployment:** Occurs due to economic cycles, where demand for labor rises or falls depending on the economic conditions.
4. **Structural Unemployment:** Happens due to changes in the structure of an industry, such as the introduction of new technologies.
5. **Frictional Unemployment:** Happens when workers are in between jobs, typically short-term in nature, as they search for new employment opportunities.

7. What is 'Seigniorage? Does it cause inflation? 97(7.b)

Seigniorage:

Seigniorage is an economic concept where the government creates new currency and circulates it in the market. It typically serves as a method of revenue generation for the government, especially when it prints new money to pay off debts or meet expenses.

Process of Inflation Creation:

1. **Increase in the Money Supply:** When the government prints new money and circulates it, the total amount of money in the market increases. For example, if the government prints 100 crore Taka, that money will circulate in the market.
2. **Relationship between Demand and Supply:** If the demand for goods and services in the market remains the same, an increase in the money supply can lead to higher prices. This is because when people have more money, they can spend more, which increases demand.
3. **Creation of Inflation:** When there is a mismatch between demand and supply (e.g., demand increases but supply remains the same), inflation may occur. In other words, more money

circulating in the market can lead to higher prices for goods and services.

4. **Qualitative Impact:** Seigniorage can cause a decrease in the value of money. The inclusion of new money can reduce people's confidence, which in turn encourages inflation.

Conclusion: Generally, seigniorage can be an economic benefit for the government, but if excessive, it can lead to inflation in the market, which can be harmful to the economy of the country.

8. According to the central bank, even if new money is printed and provided to the government it will not affect inflation much. Because a lot of money is being withdrawn by selling dollars through open market operations. Explain your opinion.

Opinion:

1. **Balance of Money Supply in the Market:** If the central bank prints new money and circulates it, it will increase the total amount of money in the market. However, by selling dollars through Open Market Operations (OMO) and withdrawing money from the market, the central bank can effectively control the money flow.
2. **Relationship between Inflation and Demand:** If the central bank withdraws money from the market in a timely manner, it may help maintain a balance between demand and supply. As a result, the possibility of inflation increasing can be reduced.
3. **Past Experience:** In some countries, even after printing new money, inflation has increased in the market. This is because printing excessive money increases people's spending tendencies, leading to a rise in the prices of goods.
4. **Impact Duration:** While the impact of printing new money might temporarily reduce inflation, if new money is not properly controlled in the long term, inflation could occur.
5. **Effect of Foreign Currency:** By selling dollars through Open Market Operations, the central bank can help increase the country's foreign currency reserves. However, this can also help stabilize the value of the currency and mitigate the effects of inflation.

Conclusion: Overall, the central bank's opinion may be reasonable, but it is a complex process where the market's response and experience are important. Without proper planning and effective management, the impact of inflation may increase.

Inflation:

Inflation is an economic condition where the general price level of goods and services increases over a specific period of time, leading to a reduction in the purchasing power of money. In other words, with the same amount of money, fewer goods or services can be purchased compared to before.

When Inflation Can Be Beneficial:

- **Tolerable Inflation:** Typically, inflation between 2% and 5% is considered tolerable and normal for an economy. This level of inflation helps stimulate economic growth and job creation.
- **Stimulative Effect:** Controlled inflation encourages consumers to spend, as they anticipate prices will rise further in the future. This helps in increasing production and investment.
- **Benefit for Borrowers:** Controlled inflation can be beneficial for borrowers, as the real value of their debt decreases over time.

Unemployment:

Unemployment refers to the condition where individuals who are capable of working and actively seeking employment are unable to find a job. This reflects an economic issue that negatively impacts the country's productivity and overall economy.

Types of Unemployment:

1. **Structural Unemployment:**

- ✓ This occurs when changes in the economy or technological advancements make previous skills irrelevant.
- ✓ Example: In Bangladesh, workers in the jute industry have faced challenges in finding new jobs due to a decline in industry demand.

2. **Frictional Unemployment:**

- ✓ This happens when an individual leaves a job or is searching for a new one, typically for a short period.
- ✓ Example: A graduate from university may experience some unemployment while searching for a job.

3. Cyclical Unemployment:

- ✓ This type of unemployment occurs during an economic downturn, when the demand for labor decreases.
- ✓ Example: During the COVID-19 pandemic in 2020, many people lost their jobs due to the economic crisis.

4. Seasonal Unemployment:

- ✓ Some industries are seasonal, meaning employment is not available year-round.
- ✓ Example: In Bangladesh, seasonal unemployment is seen in industries like mango farming and tourism.

5. Technological Unemployment:

- ✓ This occurs when technological advancements, such as automation, reduce the need for human labor.
- ✓ Example: In Bangladesh, the increase in automation in garment factories has led to some workers losing their jobs.

In Bangladesh, cyclical and structural unemployment are more prevalent, especially among the young educated population, with a high unemployment rate of 11.2%.

11. What is inflation? Why is inflation a major concern for the central banks? 99(9.b)

Inflation:

Inflation refers to the rate at which the general price level of goods and services rises over a specific period, meaning that the purchasing power of money decreases, and one can buy fewer goods with the same amount of money.

Why is it important for the Central Bank?

1. Impact on Purchasing Power:

- ✓ High inflation reduces the purchasing power of money, meaning people can buy less with the same amount of money.
- ✓ The central bank controls inflation to maintain currency stability.

2. Setting Interest Rates:

- ✓ The Bangladesh Bank can control inflation by setting interest rates.
- ✓ When inflation is high, the central bank may increase interest rates to reduce the money flow in the economy.

3. **Balancing Investment and Growth:**

- ✓ High inflation negatively affects investment.
- ✓ The central bank tries to maintain a balance between inflation and economic growth.

4. **Stability of Foreign Currency:**

- ✓ High inflation devalues the local currency and increases import costs.
- ✓ The central bank brings stability to the economy by controlling the exchange rate.

In 2023, inflation in Bangladesh surpassed 9%, which has posed a significant challenge for Bangladesh Bank.

12. How does interest rate affect inflation? Briefly explain: 99(9.c)

Relationship between Interest Rate and Inflation:

Interest rates are a key tool for controlling the flow of money in an economy. They impact inflation in the following ways:

1. **When Interest Rates Increase:**

- ✓ The cost of borrowing from banks rises, leading to a decrease in loan uptake.
- ✓ People tend to spend less and save more, causing a reduction in the money circulating in the economy.
- ✓ With decreased demand, the prices of goods decrease, meaning inflation reduces.

2. **When Interest Rates Decrease:**

- ✓ The cost of loans decreases, which boosts investment and consumption.
- ✓ More money circulates in the economy, which can lead to an increase in inflation.

The Bangladesh Bank controls interest rates to try and reduce inflation. Currently, the policy interest rate is 9%, which plays an effective role in controlling inflation. Banks need to conduct audits, verify data, and implement proper monitoring systems.

13. Define fiscal policy and monetary policy? Discuss the tools of monetary policy. 99(7.a)

Fiscal Policy:

Fiscal policy refers to the government's strategy of controlling the economy through revenue collection and expenditure. It aims to bring economic stability through taxes, government spending, subsidies, and borrowing. For example, the government of Bangladesh can stimulate the economy by increasing spending on education and infrastructure through its budget.

Monetary Policy:

Monetary policy is the strategy of influencing the economy by controlling the money supply, interest rates, and the banking system. It is regulated by the Bangladesh Bank and plays a key role in controlling inflation, investment, and economic growth.

Main Tools of Monetary Policy:

1. Policy Rate:

- ✓ The Bangladesh Bank sets the REPO and Reverse REPO rates, which influence the borrowing and investment decisions of commercial banks.

2. Cash Reserve Ratio (CRR) & Statutory Liquidity Ratio (SLR):

- ✓ **CRR (Cash Reserve Ratio):** The amount of cash commercial banks must keep with the Bangladesh Bank.
- ✓ **SLR (Statutory Liquidity Ratio):** The proportion of commercial banks' assets they must invest in government bonds and other safe assets.

3. Open Market Operations (OMO):

- ✓ The central bank buys and sells government bonds to control the money supply.

4. Foreign Exchange Policy:

- ✓ The Bangladesh Bank manages the reserves of dollars and other currencies to regulate the exchange rate.

5. Moral Suasion:

- ✓ The Bangladesh Bank provides certain recommendations to commercial banks regarding interest rates or loan distribution.

14. Describe the tools of monetary policy and fiscal policy. 96(8.a)

Monetary Policy:

Monetary policy refers to the measures taken by a central bank or monetary authority to regulate economic activities. Its primary objective is to achieve macroeconomic stability by controlling the money supply and interest rates to manage inflation, unemployment, and economic growth.

Instruments of Monetary Policy:

1. Interest Rate Adjustment:

- ✓ The central bank increases or decreases the reference interest rate to control the money supply.
- ✓ A higher interest rate reduces borrowing, decreasing the money supply and helping control inflation.

2. **Cash Reserve Ratio (CRR):**

- ✓ Banks are required to maintain a certain percentage of their deposits as reserves with the central bank.
- ✓ Increasing the CRR reduces banks' lending capacity, thereby decreasing the money supply.

3. **Statutory Liquidity Ratio (SLR):**

- ✓ Banks must hold a specific proportion of their assets in liquid form.
- ✓ A higher SLR restricts banks' ability to lend, reducing the money supply.

4. **Open Market Operations (OMO):**

- ✓ The central bank buys or sells government securities in the open market to regulate the money supply.
- ✓ Purchasing securities increases the money supply, while selling them reduces it.

Fiscal Policy:

Fiscal policy refers to government measures aimed at regulating economic activities through revenue collection and public expenditure. Its primary goal is to achieve economic stability by managing taxation and government spending.

Instruments of Fiscal Policy:

1. **Taxation System:**

- ✓ The government collects revenue through various taxes, such as income tax, VAT, and corporate tax.
- ✓ Changing tax rates influences economic activities.

2. **Government Spending:**

- ✓ The government invests in key sectors like education, healthcare, and infrastructure to stimulate economic growth.
- ✓ Increased government spending boosts demand and drives economic activity.

3. **Economic Stimulus:**

- ✓ During economic downturns, the government provides financial incentives to encourage economic growth.

4. Subsidies:

- ✓ The government offers subsidies in various sectors to lower the price of essential goods, making them more affordable for the public.

15. Discuss the recent changes in monetary policy issued by Bangladesh Bank. 98(7.b)

Bangladesh Bank announces its monetary policy every six months, applicable for the January-June and July-December periods. The recent monetary policy has placed special emphasis on controlling inflation. In October 2024, Bangladesh Bank increased its key interest rate by 50 basis points to 10%, effective from October 27. This marks the fourth interest rate hike this year. The decision was driven by inflationary pressures caused by political instability and challenges in the garments industry.

Although overall inflation slightly declined in September, food inflation remains above 12%, significantly impacting the country's 170 million people. Bangladesh Bank Governor Ahsan H. Mansur has expressed hope that inflation will decrease significantly next year, but interest rates may remain high for some time. To restore economic stability, the interim government has sought \$5 billion in financial assistance from international lenders. Since the Ukraine war in 2022, Bangladesh has already received \$4.7 billion in support from the IMF to tackle economic challenges.

In the recent monetary policy, Bangladesh Bank has prioritized inflation control and aims to maintain economic stability by increasing interest rates. Additionally, various measures have been taken to address the dollar crisis and strengthen foreign exchange reserves. The central bank has outlined strategies in its new monetary policy to mitigate the dollar shortage, according to relevant sources.

16. Considering the current economic situation, which one do you suggest; expansionary or contractionary monetary policy. Explain with logic,97(9.b)

Selection of Monetary Policy:

Based on the current economic situation, I support adopting an expansionary monetary policy. The reasons behind this are as follows:

1. **Economic Growth Enhancement:** Expansionary monetary policy helps boost investment and spending by lowering interest rates. This leads to business growth and job creation, positively impacting the country's economic growth.
2. **Reduction in Unemployment:** Lower interest rates facilitate employment growth, as establishing and expanding new businesses becomes easier. This will help reduce the unemployment rate.

3. **Encouraging New Entrepreneurs:** The availability of loans on easy terms will be encouraging for new entrepreneurs, contributing to industrial and economic development in the country.
4. **Inflation Control:** When economic growth is stable, inflation can be kept under control. Expansionary policies may temporarily increase inflation, but they are generally beneficial for the overall economy.
5. **Support in Currency Regulation:** Increased money supply in the market helps boost spending, keeping the supply chain active.

Therefore, an expansionary monetary policy should be adopted in the current economic situation to ensure the country's economic development and stability.

17. Define National Income. Explain the different methods of computing National Income. 98(8.a)

National Income:

National income is the total market value of all final goods and services produced in a country within a specific period (usually one year). It measures the overall economic activity of a country and serves as an indicator of economic development.

There are three main methods of measuring national income:

1. **Production Method:** This method calculates national income by summing the total value of final goods and services produced in different sectors of the economy. Only final goods and services are considered to avoid double counting.
2. **Income Method:** This method determines national income by adding the earnings from factors of production (labor, capital, land, and entrepreneurship), such as wages, profits, rent, and interest.
3. **Expenditure Method:** In this method, national income is measured by summing the total expenditure on final goods and services in different sectors of the economy. It is generally divided into four components: consumer expenditure, investment expenditure, government expenditure, and net exports (exports minus imports).

The values obtained through these three methods should theoretically be equal, as they measure the same economic activity from different perspectives.

18. Define National Income. Explain the methods of computing national income. Do you think household services should include in national income? 96(7.a)

National Income:

National income is an indicator of a country's economic activity that represents the total value of goods and services produced within a specific period (usually one year). It measures the country's economic development, growth, and overall well-being. National income is generally measured using different methods, such as Gross Domestic Product (GDP), Gross National Product (GNP), and Net National Product (NNP).

Methods of Measuring National Income:

1. **Production Method:** This method measures national income by calculating the total value of goods produced through all production activities in a country. It sums up the total value of goods and services produced in all industries and sectors.
2. **Income Method:** This method calculates national income by summing up all earnings generated through production, including wages, profits, rent, and interest.
3. **Expenditure Method:** This method measures national income by summing the total expenditure within the country, including personal consumption, government spending, investment, and net exports (exports minus imports).

Opinion on Including Household Services:

I believe that household services should be included in national income measurement. Household services, such as domestic work, child care, and elderly care, are an essential part of a country's economic activities. Although these services do not involve direct monetary transactions in the economic market, they are highly valuable and play a crucial role in societal well-being.

19. Discuss the causes of current unfavorable Balance of Payment (BoP) situation of Bangladesh. What steps do you suggest to solve the current BoP crisis. 97(9.c)

Causes of Bangladesh's BOP Situation:

1. **Trade Deficit:** Bangladesh faces a trade deficit as import expenses exceed export earnings. The high import of essential and advanced products has worsened the situation.
2. **Decline in Remittances:** The inflow of remittances from Bangladeshis working abroad is decreasing, which is a crucial source of foreign currency earnings.

3. **Inflation:** High inflation is increasing domestic production costs, making exports more expensive and reducing the competitiveness of Bangladeshi products in international markets.
4. **Global Economic Pressure:** Rising fuel prices in the international market and political instability have negatively impacted Bangladesh's economy, leading to a decline in foreign investment.
5. **Resulting Adverse Conditions:** Due to the above reasons, foreign currency reserves are depleting, posing risks to the country's economic stability.

Steps to Resolve the BOP Crisis:

1. **Boosting Exports:** The government should explore new export markets and diversify export products. Improving product quality and adhering to international standards is essential.
2. **Enhancing Local Production:** To reduce import dependency, local production should be increased. Government policies should support local industries and encourage innovation.
3. **Increasing Remittances:** The safety and benefits of migrant workers should be improved to encourage them to send more remittances. The government should take steps to expand access to foreign labor markets and enhance cooperation.
4. **Economic Reforms:** Macroeconomic stability should be ensured through reforms in monetary and fiscal policies, which will help attract investment.
5. **Addressing Climate Change:** Measures should be taken to mitigate the impact of climate change on agriculture and production, reducing the economic damage caused by climate-related issues.
6. **International Cooperation:** Strengthening ties with international organizations will help attract foreign investment and financial assistance to tackle the economic crisis.

Conclusion:

To overcome Bangladesh's BOP crisis, the government must take integrated and effective measures to restore economic stability and ensure overall development.

Balance of Payments (BoP):

The Balance of Payments (BoP) is a detailed record of a country's international financial transactions, reflecting the flow of foreign transactions over time. It is primarily divided into three sections:

1. **Current Account:** Includes exports and imports of goods and services, income, and transfers.
2. **Capital Account:** Records foreign investments and government loans.
3. **Financial Account:** Includes foreign bank transactions and investments.

Reasons Why the Balance of Payments Always Balances:

1. **Accounting Accuracy:** The BoP always balances because it is a comprehensive record of all financial transactions within a specific period, ensuring equilibrium.
2. **Necessary Accounting Principle:** If there is a deficit in the current account, there will be a corresponding surplus in the capital account, maintaining the overall balance.
3. **Basic Economic Theory:** According to fundamental economic principles, if a country's imports increase, it can compensate for the deficit through foreign investment or external borrowing, ensuring balance.
4. **Economic Policies:** The government and central bank implement various policies, such as monetary and fiscal measures, to maintain the balance of international transactions.

Module-G: Macroeconomic Performance of Bangladesh

1. Define the differences between Gross National Product (GNP) and Gross Domestic Product (GDP). 96(7.b)

Gross National Product (GNP):

GNP refers to the total monetary value of all goods and services produced by a country's citizens and businesses over a specific period, regardless of their location. It includes the income earned by nationals living abroad but excludes the income generated by foreigners within the country.

Gross Domestic Product (GDP):

GDP represents the total monetary value of all goods and services produced within a country's geographical boundaries, regardless of who produces them. It includes the production of foreign companies operating within the country but excludes the income earned by the country's citizens abroad.

Differences between GNP and GDP:

1. **Focus Difference:** GNP emphasizes the total income of a country's citizens, whereas GDP focuses on domestic production within the country's borders.
2. **Income Inclusion:** GNP includes the income of citizens living abroad, while GDP excludes it.
3. **Foreign Income Consideration:** GDP includes the income earned by foreigners within the country, whereas GNP does not.

2. Define Gross Domestic Product (GDP) and Gross National Product (GNP). 98(9.b)

Gross Domestic Product (GDP):

The total market value of all final goods and services produced within a country's geographical boundaries over a specific period (usually one year) is called Gross Domestic Product (GDP).

Formula:

$$\text{GDP} = \text{Consumption (C)} + \text{Investment (I)} + \text{Government Spending (G)} + \text{Net Exports (X - M)}$$

Gross National Product (GNP):

The total market value of all goods and services produced by a country's citizens over a specific period, whether produced domestically or abroad, is called Gross National Product (GNP).

Formula:

$$\text{GNP} = \text{GDP} + \text{Income earned from abroad} - \text{Income earned by foreigners within the country}$$

Key Differences:

- **GDP** includes only production within the country.
- **GNP** includes the total production of a country's citizens, whether inside or outside the country.

3. Explain the types of unemployment.

Types of Unemployment

Unemployment occurs when a working-age person is unable to find a job. Unemployment is classified into various categories in economics. The main types are:

1. **Structural Unemployment:** This occurs when changes in the structure of the labor market lead to a decrease in demand for workers with specific skills. For example, some jobs become obsolete due to technological advancements.
2. **Frictional Unemployment:** This is short-term unemployment that occurs when a person transitions from one job to another. It can happen when new graduates search for jobs or when individuals make changes in their careers.
3. **Cyclical Unemployment:** This occurs during economic recessions when businesses reduce production and lay off workers. This type of unemployment decreases when the economy recovers.
4. **Seasonal Unemployment:** This occurs when the demand for labor decreases during specific seasons. For example, seasonal unemployment can be found in agriculture, tourism, and construction industries due to seasonal work patterns.
5. **Disguised Unemployment:** This occurs when a person is formally employed but their productivity is either nonexistent or very low. It is commonly seen in agriculture and informal sectors where excessive workers are employed but their contribution to production is minimal.

Unemployment is a significant issue for the economy, as it can lead to reduced productivity and income. Therefore, the government takes various measures to reduce unemployment.

4. Discuss the General Characteristics of Macroeconomic Accounts.

General Characteristics of Macroeconomic Accounts

Macroeconomic accounts are a framework used to measure the overall activities of an economy, which is used to analyze various economic indicators. The main characteristics are:

1. **Measurement of Overall Economic Activity:** It compiles data on production, expenditure, income, and investment over a specific period, helping to understand the overall state of the economy.
2. **National Income Accounting:** Macroeconomic accounts determine GDP, GNP, National Income (NI), Net National Income (NNI), and the levels of expenditure and savings, which are important for measuring economic growth and stability.
3. **Sectoral Analysis:** It analyzes different sectors of the economy, such as agriculture, industry, services, and government spending, which aids in policy formulation.
4. **Inclusion of International Transactions:** It tracks imports, exports, and foreign investments, which helps determine the balance of foreign transactions.
5. **Assistance in Policy Formulation:** Macroeconomic accounts play a crucial role in formulating economic policies, such as monetary policy and fiscal policy.

These characteristics help in analyzing macroeconomics and policy formulation, contributing to maintaining economic stability.

Difference:

1. Inflation vs. Deflation

- ✓ **Inflation:** Refers to the continuous rise in the general price level in the economy, which reduces the purchasing power of money.
- ✓ **Deflation:** The opposite process where the overall price level decreases, leading to an increase in the purchasing power of money.
- ✓ **Key Difference:** Inflation occurs due to excess demand in the economy, while deflation typically happens during a recession or low demand.

2. Inflation vs. Relative Price Changes

- ✓ **Inflation:** The rise in the average price of all goods due to the increase in the general price level.
- ✓ **Relative Price Changes:** The price of a specific good changes in relation to other goods, depending on market supply and demand.
- ✓ **Key Difference:** Relative price changes refer to the price changes of specific goods, while inflation refers to the change in the overall price level of the economy.

3. Disinflation vs. Deflation

- ✓ **Disinflation:** A reduction in the rate of inflation, but it remains positive (e.g., from 7% to 4%).
- ✓ **Deflation:** Occurs when the inflation rate turns negative, meaning prices decrease.

- ✓ **Key Difference:** In disinflation, the rate of price increase slows down, but in deflation, prices actually decrease.

4. **Headline Inflation vs. Core Inflation**

- ✓ **Headline Inflation:** Measures the overall inflation rate including volatile price components like food and fuel.
- ✓ **Core Inflation:** Measures the inflation rate excluding food and fuel, as their prices tend to fluctuate widely.
- ✓ **Key Difference:** Core inflation is more effective for analyzing long-term trends, while headline inflation is used to understand short-term market conditions.

Short Notes:

1. Economic Systems

An economic system refers to the structure and methods by which a society organizes the production, distribution, and consumption of goods and services. There are four main types of economic systems:

- **Market Economy:** Decisions are made by individuals based on supply and demand.
- **Command Economy:** The government controls the production and distribution of goods.
- **Mixed Economy:** Combines elements of both market and command economies.
- **Traditional Economy:** Based on customs and traditions with limited changes in the production system. Each system has advantages and disadvantages, depending on the society's goals for economic efficiency, equality, and individual freedoms.

2. Consumer Behaviour

Consumer behavior is the study of how individuals make decisions to spend their resources (time, money, effort) on consumption-related items. It is influenced by psychological, social, and economic factors. Consumers' preferences are influenced by income, tastes, preferences, and the prices of goods and services. It also involves understanding the decision-making process regarding what, when, and how much to purchase. Consumer behavior is essential for businesses to design marketing strategies and optimize product offerings.

3. Marginal Rate of Substitution (MRS)

The Marginal Rate of Substitution (MRS) refers to the rate at which a consumer is willing to exchange one good for another while maintaining the same level of satisfaction or utility. It is represented by the slope of the indifference curve. As a consumer substitutes one good for another, the MRS typically decreases, reflecting diminishing marginal utility. The MRS is crucial in understanding consumer preferences and in deriving demand curves.

4. Budget Line

A budget line represents all possible combinations of two goods that a consumer can purchase given their income and the prices of those goods. It shows the trade-offs a consumer faces when deciding how to allocate their income between different goods. The slope of the budget line is determined by the relative prices of the goods. Any point on the budget line shows an affordable combination, while any point inside the line is a feasible but inefficient allocation.

5. Price Elasticity of Demand

Price elasticity of demand measures the responsiveness of the quantity demanded of a good to a change in its price. It is calculated as the percentage change in quantity demanded divided by the percentage change in price. If demand is elastic, consumers are highly responsive to price changes, while inelastic demand indicates less responsiveness. The elasticity helps businesses set prices and predict consumer behavior in response to price changes.

6. Perfectly Elastic

Perfectly elastic demand occurs when any change in the price of a good results in an infinite change in the quantity demanded. This means that at a specific price, consumers are willing to buy any quantity of the good, but they will buy none if the price changes, no matter how small. Perfectly elastic demand is a theoretical extreme, rarely seen in real life, and is represented by a horizontal demand curve.

7. Perfectly Inelastic

Perfectly inelastic demand is when the quantity demanded of a good does not change regardless of any change in its price. This means that consumers will continue to purchase the same amount of the good even if the price increases or decreases. It is represented by a vertical demand curve. An example could be life-saving medications where consumers must buy the same amount regardless of price changes.

8. Isoquant Curves

Isoquant curves represent all the combinations of two inputs (such as labor and capital) that produce the same level of output. It is similar to an indifference curve but for production instead of consumption. The curve is downward sloping and convex to the origin, reflecting diminishing marginal returns. The shape of the isoquant helps businesses understand the trade-off between different production inputs and the efficiency of their use.

9. Implicit Cost

Implicit costs refer to the opportunity costs of using resources owned by the business owner, where no direct payment is made. These costs are not reflected in accounting profits but represent the foregone income from not using those resources in the next best alternative. For example, the owner's time spent managing the business could have been used in another income-generating activity. Implicit costs are crucial in calculating economic profits.

10. Explicit Cost

Explicit costs are actual out-of-pocket expenses incurred by a business in its production process. These are direct, measurable costs such as wages, rent, and material expenses. Explicit costs are recorded in the financial statements and affect the accounting profits of a business. They are different from implicit costs, which are not recorded in the accounts but still represent real economic costs. Explicit costs are essential in understanding the financial health of a business.

11. Profit Maximization

Profit maximization is the process by which firms determine the price and output level that returns the greatest profit. This occurs when the marginal cost (MC) of producing an additional unit of output equals marginal revenue (MR). Firms aim to produce at the level where $MC = MR$, ensuring they are not overproducing (which would increase costs) or underproducing (which would mean missed opportunities for profit). This concept is central in microeconomics and helps businesses make decisions that maximize their profitability.

12. Productive Inefficiency

Productive inefficiency occurs when an economy or firm produces goods or services at a higher cost than necessary. This often happens when resources are not utilized in the most effective manner, meaning that output could be produced with fewer inputs or at a lower cost. In a productive inefficiency scenario, firms are not operating at the lowest point of their average cost curve, and this reduces overall economic welfare. It typically leads to waste and higher prices for consumers.

13. Allocative Inefficiency

Allocative inefficiency happens when resources are not allocated in the most beneficial way to maximize societal welfare. It occurs when the price of a good or service does not reflect its marginal cost ($P \neq MC$). In perfect competition, allocative efficiency is achieved when firms set the price equal to the marginal cost of production. When allocative inefficiency occurs, it results in the overproduction or underproduction of certain goods, leading to a misallocation of resources.

14. Market Intervention

Market intervention refers to the actions taken by the government or regulatory bodies to influence the functioning of markets. This can involve setting price floors (minimum prices), price ceilings (maximum prices), tariffs, quotas, subsidies, and regulations. The aim of market intervention is often to correct market failures, ensure fair competition, or protect consumers and producers from excessive price fluctuations. However, interventions can sometimes lead to inefficiencies and unintended consequences.

15. Income Distribution

Income distribution refers to the way in which a nation's total income is distributed among its population. It is influenced by factors such as wages, investments, taxation, and government transfers. Inequalities in income distribution can lead to social and economic problems, including poverty, inequality, and reduced access to education and healthcare. Governments often implement policies to redistribute income more evenly through progressive taxation, welfare programs, and social security.

16. Progressive Taxes

Progressive taxes are tax systems where the rate of taxation increases as the income of the taxpayer rises. This means that individuals with higher incomes pay a higher percentage of their income in taxes than those with lower incomes. Progressive taxes aim to reduce income inequality by redistributing wealth from wealthier individuals to those with lower incomes. Common examples include income taxes, where tax rates increase in steps as income rises.

17. Regressive Taxes

Regressive taxes are taxes where the tax rate decreases as the income of the taxpayer increases. In other words, lower-income individuals pay a higher percentage of their income in taxes compared to wealthier individuals. Sales taxes and value-added taxes (VAT) are examples of regressive taxes, as they take a larger proportion of income from lower-income individuals, who spend a larger share of their earnings on goods subject to these taxes.

18. Proportional Taxes

Proportional taxes, also known as flat taxes, are tax systems in which the tax rate remains constant regardless of the taxpayer's income. Everyone pays the same percentage of their income in taxes. For example, a flat tax of 10% would mean that both a person earning \$30,000 and a person earning \$300,000 would pay 10% of their income in taxes. This type of tax is simpler to administer but may not reduce income inequality as effectively as progressive taxes.

19. Transfer Payments

Transfer payments are payments made by the government to individuals without any goods or services being provided in return. These payments are typically aimed at reducing inequality or providing financial support to vulnerable groups, such as the elderly, unemployed, or low-income individuals. Examples include unemployment benefits, pensions, and welfare payments. Transfer payments help in income redistribution and support individuals during times of economic hardship.

20. Aggregate Demand

Aggregate demand (AD) represents the total quantity of goods and services demanded in an economy at a given overall price level and within a specific period. It is the sum of consumption, investment, government spending, and net exports (exports minus imports). The aggregate demand curve typically slopes downward, meaning that as the price level falls, the quantity of goods and services demanded increases. Changes in any of the components of AD can lead to shifts in the curve, affecting overall economic output.

21. Aggregate Supply

Aggregate supply (AS) is the total quantity of goods and services that producers in an economy are willing and able to produce at different price levels in a given time period. The aggregate supply curve can be vertical, horizontal, or upward sloping depending on the time horizon (short-run or long-run). In the short-run, AS can be influenced by factors like input prices and wage rates, while in the long-run, it is determined by factors such as technological advancements and capital stock.

22. Business Cycle

The business cycle refers to the fluctuations in economic activity that an economy experiences over time, typically consisting of four phases: expansion, peak, contraction (recession), and trough. During expansion, economic output increases, leading to higher employment and growth. The peak represents the point of maximum economic output, after which a contraction occurs, resulting in a decline in economic activity. The trough marks the lowest point, after which the cycle may begin to expand again.

23. Supply-side Policies

Supply-side policies are aimed at increasing the overall supply of goods and services in an economy. These policies focus on improving the efficiency and productivity of production by reducing barriers to production, such as high taxes, excessive regulation, and labor market restrictions. Examples include tax cuts for businesses, deregulation, and policies to improve education and workforce skills. The goal of supply-side policies is to increase long-term economic growth by encouraging investment and production.

24. Fiscal Accounts

Fiscal accounts refer to the record of a government's revenue and expenditure, which reflects its fiscal policy. These accounts include tax revenues, government spending on goods and services, and transfers. Fiscal accounts are essential for assessing the government's financial health and sustainability. A fiscal surplus occurs when government revenues exceed expenditures, while a fiscal deficit occurs when expenditures surpass revenues. Fiscal policies can influence national savings, investment, and overall economic stability.

25. Monetary Accounts

Monetary accounts track the supply and demand for money within an economy. This includes the central bank's management of monetary policy, interest rates, and the money supply. The monetary accounts aim to control inflation, stabilize the currency, and promote economic growth. Key tools used include setting reserve requirements for banks, engaging in open market operations, and adjusting the benchmark interest rates. Monetary policy works in tandem with fiscal policy to manage an economy's performance.

26. Per Capita GDP

Per capita GDP is a measure of the average economic output per person in a country. It is calculated by dividing a country's total Gross Domestic Product (GDP) by its population. This indicator helps to compare the standard of living between countries or regions. While it provides a useful snapshot of average income levels, it does not account for income inequality or differences in cost of living across regions.

27. GNI (Gross National Income)

Gross National Income (GNI) is the total income earned by a country's residents both domestically and internationally, including income earned by citizens working abroad or through investments. It is the sum of GDP and net income from abroad (such as income from foreign investments). GNI is a broader measure of a country's economic health than GDP, as it includes international income flows. It is often used to assess the economic well-being of a nation and to compare economies globally.